



Time series analysis and forecasting with applications in climate science

Gnanashree Mysuru Venkatesh

December 2025

School of Mathematics,
Cardiff University

A dissertation submitted in partial fulfilment of the
requirements for MSc (in Data Science and Analytics) by taught programme,
supervised by Prof. Andrey Pepelyshev

APPENDIX 1: Specimen Layout for Declaration/Statements page to be included in Taught Master's Degree Dissertations

CANDIDATE'S ID NUMBER	C24051159
CANDIDATE'S SURNAME	Miss Mysuru
CANDIDATE'S FULL FORENAMES	Gnanashree Venkatesh

DECLARATION

This work has not previously been accepted in substance for any degree and is not concurrently submitted in candidature for any degree.

Signed Gnanashree Mysuru Venkatesh (candidate) Date ...15/12/2025.....

STATEMENT 1

This dissertation is being submitted in partial fulfillment of the requirements for the degree of ...MSc... (insert MA, MSc, MBA, MScD, LLM etc, as appropriate)

Signed Gnanashree Mysuru Venkatesh (candidate) Date ...15/12/2025.....

STATEMENT 2

This dissertation is the result of my own independent work/investigation, except where otherwise stated. Other sources are acknowledged by footnotes giving explicit references. A Bibliography is appended.

Signed Gnanashree Mysuru Venkatesh (candidate) Date ...15/12/2025.....

STATEMENT 3 – TO BE COMPLETED WHERE THE SECOND COPY OF THE DISSERTATION IS SUBMITTED IN AN APPROVED ELECTRONIC FORMAT

I confirm that the electronic copy is identical to the bound copy of the dissertation

Signed Gnanashree Mysuru Venkatesh (candidate) Date ...15/12/2025.....

STATEMENT 4

I hereby give consent for my dissertation, if accepted, to be available for photocopying and for inter-library loan, and for the title and summary to be made available to outside organisations.

Signed Gnanashree Mysuru Venkatesh (candidate) Date ...15/12/2025.....

STATEMENT 5 - BAR ON ACCESS APPROVED

I hereby give consent for my dissertation, if accepted, to be available for photocopying and for inter-library loans **after expiry of a bar on access approved by the Graduate Development Committee.**

Signed Gnanashree Mysuru Venkatesh (candidate) Date ...15/12/2025.....

Acknowledgements

I would like to express my sincere gratitude to my internal supervisor, **Prof. Andrey Pepelyshev**, for being readily available whenever I had queries and for arranging meetings to provide guidance throughout the course of this research. His continuous support, constructive feedback, and encouragement have been invaluable in shaping the direction of this work.

...

Finally, I would like to thank my peers in postgraduate study and the **School of Mathematics** for its academic support and for fostering an excellent research environment in which this work was carried out here.

Executive Summary

Aim of the study

The objective of the current dissertation was the examination of the long-term climate dynamics by the use of efficient time series analysis approaches, specifically focusing on the behaviour of the major climate variables for the last 50-70 years. For the analysis, the Singular Spectrum Analysis (SSA) technique was employed on the major climate time series:

- CO2 concentration, atmospheric
- Enso Index of El Niño-Sou
- Global Surface Temperature Anomalies
- Sea Ice Cover Extent

Short-term forecasting was subsequently carried out using ARIMA on the SSA-reconstructed components.

Datasets used: CO₂, ENSO, Global Temperature, Arctic Sea Ice

The analysis employed long, high-quality monthly data sets provided by worldwide scientific organisations (NASA, NOAA, NSIDC, Met Office):

- CO2: Dataset of the Mauna Loa Observatory (1958-2023)
- ENSO: NOAA Niño 3.4 monthly index (1950-2023)
- Global Temperature: NASA GISTEMP monthly anomalies (1880–2023, subset used:1950-2023)
- Arctic Sea Ice: NSIDC Sea Ice Index (1979-2023)

Each data set corresponds to a distinct characteristic of climate system behaviour, ranging from greenhouse gases, ocean-atmosphere interaction, planetary warming, to polar ice caps.

Method: Singular Spectrum Analysis (SSA) + Forecasting

SSA was employed as the main analysis method because it entails the following:

- Trend, Seasonal, and Multi-year Cycle Component Decomposition
- long-term climate signal extraction, without the assumption of linearity
- clear reconstruction of key components
- better forecasting when used together with existing models

The methodology involved the following steps:

- 1) Embedding: Every time series is transformed into a trajectory matrix.
- 2) SVD Decomposition: To split the data into eigen-components.
- 3) Grouping: Trend, Seasonality, and Cyclic Components
- 4) Reconstruction: Combine the components above to get clean signals.
- 5) Forecasting: Use ARIMA on the SSA-trend or SSA-reconstructed time series.

This technique helps suppress the effects of noise and provides more meaningful results regarding climate patterns compared to classical time series analysis.

Key findings from all four datasets

1. Atmospheric CO₂ — Strong, accelerating upward trend

- SSA found that the series exhibited a smooth increasing trend with strong annual cycles.
- CO₂ concentrations accelerated around 2000, typical of the growth of industry.
- Estimates predict that the concentration of CO₂ will keep rising in the short term, given the current rate of increase.

2. ENSO Variability - Strong multi-year oscillations

- Clear patterns of oscillation of 2-7 years were identified (El Niño/La Niña cycles).
- ENSO did not demonstrate any positive or negative trend, proving that it is a natural climate variability indicator, not a warming force.
- ENSO affects short-term variations of global temperature.

3. Global Temperature: The steady warming trend

- SSA trend: steady warming since about 1970.
- The warming starts accelerating sharply after 1998.
- The variation for the seasonal component is smaller.
- Projections also see a continued warming.

4. Arctic Sea Ice: Fast Reduction

- SSA demonstrated a distinct, steady decline since 1979.
- Even the cyclical patterns persist, although the downtrend clearly prevails.
- Decline accelerates after ~2005
- Forecast: Continued reduction, indicating loss of summer sea ice.

Practical interpretation of results

The results of the four climate data sets provide a consistent and scientifically valid picture of climate change:

- The increasing level of CO₂ is the clearest and most consistent trend. This is verified by the SSA, showing steady growth since the 1960s, caused by the emission of CO₂ by fossil fuels, industry, and changes in land use. The annual cycle mirrors the natural processes of photosynthesis, but does not neutralise the increasing level.
- Global temperature follows the same path as CO₂. Analysis of the SSA decomposition of the temperature data also shows that the warming of the Earth's temperature starts accelerating around the end of the 20th century, consistent with the acceleration of CO₂. This only adds strength to the already fact that the warming of the planet is a result of greenhouse gases.
- Even though the effects of ENSO on short-term climate and weather patterns are enormous, there is no indication of any climate shift. The cycles of SSA indicate that the periodicities of 2-7 years behave around zero. This is clear evidence that the impacts of ENSO affect climate variability rather than climate change.
- The Extent of the Sea Ice in the Arctic Ocean remotely reflects the clearest indication of climate change. The SSA data sets strongly identify the steady and increasing melt of the ice, especially since the year 2000. The melt of sea ice is very well connected with the sensation of rising Earth temperatures.
- Taken together, the four data sets provide a consistent sequence of climate-related phenomena: CO₂ levels go up → Warming occurs → Reductions of Arctic ice, being

influenced by El Niño-Southern Oscillation cycles, with ENSO adding short-term variability but not affecting the long-term trajectory.

- Forecasts produced by the use of the ARIMA model on the SSA reconstructed trends indicate the
 - 1) CO₂ levels increase
 - 2) global temperature increase
 - 3) Arctic sea ice loss
 - 4) and periodic cycles of ENSO that will keep on temporarily intensifying and modulating Global Climate Events.

These findings reinforce the current scientific view that the main cause of current climate change is the emission of greenhouse gases generated by human activity, and that natural cycles, although inducing variability on the short time scales of ENSO, do not affect the long-term warming trend.

Table of Contents

Acknowledgements.....	3
Executive Summary.....	4
List of Acronyms.....	7
Table of Contents.....	8
List of Figure.....	9
CHAPTER 2 — LITERATURE REVIEW.....	13
2.1 SSA Applications in Climate Studies.....	13
2.2 Atmospheric CO ₂ Trend Studies.....	13
2.3 ENSO Variability Studies.....	13
2.4 Global Temperature Trend Literature.....	14
2.5 Arctic Sea Ice Decline Research.....	14
2.6 Research Gaps.....	14
CHAPTER 3 — DATA & PREPROCESSING.....	16
3.1 CO ₂ Dataset Description + Raw Time-Series Plot.....	16
3.2 ENSO Dataset Description + Raw Time-Series Plot.....	17
3.3 Global Temperature Dataset + Raw Time-Series Plot.....	17
3.4 Arctic Sea Ice Dataset + Raw Time-Series Plot.....	18
3.5 Data Cleaning & Time-Series Conversion.....	19
3.5.1: Missing Values.....	19
3.5.2: Dataset Standardisation.....	19
3.5.3: Dataset Standardisation.....	19
3.5.4: SSA Readiness.....	20
CHAPTER 4 — METHODOLOGY.....	20
4.1 Overview of Singular Spectrum Analysis.....	20
4.2 Embedding Dimension Selection.....	20
4.3 Singular Value Decomposition (SVD).....	21
4.4 Eigenvectors & Grouping Strategy.....	22
4.5 Component Reconstruction.....	22
4.6 Forecasting Method (ARIMA on SSA Components).....	23
4.7 Software Implementation in R.....	23
CHAPTER 5 — SSA RESULTS.....	24
5.1 CO ₂ SSA Trend, Seasonal & Residual Components.....	24
5.1.1 Scree Plot and Eigenvalue Interpretation.....	24
5.1.2 Eigenvalue Interpretation.....	25

5.1.3 Seasonal Component Reconstruction.....	25
5.1.4 Multi-Year Oscillatory Components.....	26
5.1.5 Full SSA Reconstruction.....	27
5.2 ENSO SSA Components.....	28
5.2.1 Scree Plot and Eigenvalue Spectrum.....	28
5.2.2 Eigenvector Patterns and Component Interpretation.....	28
5.2.3 Component Reconstruction.....	29
5.2.4 Residual Analysis.....	30
5.2.5 Interpretation.....	30
5.3 Global Temperature SSA Components.....	31
5.3.1 Scree Plot and Eigenvalue Structure.....	31
5.3.2 Eigenvector Interpretation.....	32
5.3.3 Trend Reconstruction.....	32
5.3.4 Oscillatory Components (ENSO-like + Decadal Modes).....	33
5.3.5 Reconstruction vs Original Series.....	34
5.4 Arctic Sea Ice SSA Components.....	34
5.4.1 Scree Plot and Eigenvalue Pattern.....	34
5.4.2 Eigenvector Interpretation.....	35
5.4.3 Trend Component (Long-Term Decline).....	36
5.4.4 Seasonal Component.....	36
5.4.5 Multi-Year Variability.....	37
5.4.6 Reconstruction.....	38
CHAPTER 6 — FORECASTING RESULTS.....	39
6.1 Forecasting Framework.....	39
6.2 SSA–ARIMA Forecasts for Atmospheric CO ₂	39
6.2.1 Model Setup.....	39
6.2.2 Forecast Results.....	40
6.3 SSA–ARIMA Forecasts for ENSO (ONI).....	40
6.3.1 Model Setup.....	40
6.3.2 Forecast Results.....	41
6.4 SSA–ARIMA Forecasts for Global Temperature Anomalies.....	41
6.4.1 Model Setup.....	41
6.4.2 Forecast Results.....	42
6.5 SSA–ARIMA Forecasts for Arctic Sea Ice Extent.....	42
6.5.1 Model Setup.....	42
6.5.2 Forecast Results.....	43
6.6 Comparative Assessment of Forecasts.....	43

CHAPTER 7 — DISCUSSION.....	44
7.1 CO ₂ and Temperature Relationship.....	44
7.2 ENSO Influence on Climate Variability.....	45
7.3 Temperature vs Arctic Sea Ice Decline.....	45
7.4 Physical Interpretation of Results.....	45
7.5 Policy & Climate Implications.....	46
CHAPTER 8 — LIMITATIONS.....	47
8.1 Data Quality Limitations.....	47
8.2 SSA Window Size Sensitivity.....	47
8.3 Forecast Uncertainty.....	47
8.4 Model Assumptions.....	48
CHAPTER 9 — CONCLUSION & FUTURE WORK.....	48
9.1 Summary of key achievements.....	48
9.2 Main climate insights.....	48
9.3 Future improvements.....	49
9.4 Possible PhD / further research direction.....	49
Chapter 6 – Discussion and Critical Evaluation.....	51
6.1 Reflection on the Use of Data Journalism in Health Communication.....	51
6.2 Assessing the Bee Narrative as a Tool for Engagement.....	51
6.3 Strengths and Weaknesses of the Approach.....	51
6.4 Alternative Visualisation or Narrative Techniques.....	51
6.5 Implications for Future Campaigns by Welsh Blood Service.....	51
6.6 Recommendations for Integrating Stem Cell Registry Promotion.....	51
Chapter 7 – Conclusion.....	51
7.1 Summary of Findings.....	51
7.2 Contribution to Welsh Blood Service’s Outreach Goals.....	52
7.3 Significance of Data Journalism in Public Health Awareness.....	52
7.4 Recommendations for Future Work and Collaboration.....	52
7.5 Final Reflection on Community and Collective Action.....	52
References.....	52

Table of Contents

Figure 1.1 Monthly Atmospheric CO ₂ Concentration at Mauna Loa(raw plot).....	19
Figure 2 Monthly ENSO Index (SOI) Time Series.....	20
Figure 3 Global Temperature Anomaly (Monthly, 2003–2024).....	21
Figure 4 Monthly Arctic Sea Ice Extent (NSIDC G02135).....	22
Figure 5 SSA Scree Plot for CO ₂ Time Series.....	28
Figure 6 SSA Eigenvectors (First 12 Components) for CO ₂	29
Figure 7 SSA Trend Component of Atmospheric CO ₂ Smooth monotonic component representing the long-term increase in atmospheric CO ₂ concentrations.....	30
Figure 8 SSA Seasonal Component of Atmospheric CO ₂ (Annual Cycle).....	31
Figure 9 Residual After removing SSA signal.....	32
Figure 10 CO ₂ Time Series: Original vs SSA Reconstruction.....	32
Figure 11 SSA Scree Plot for ENSO Index.....	33
Figure 12 SSA Eigenvectors (First Components) for ENSO Eigenvectors illustrating oscillatory structure used to guide component grouping.....	34
Figure 13 SSA Periodic/Oscillatory Component of ENSO Multi-frequency oscillatory behaviour associated with ENSO variability.....	35
Figure 14 ENSO Time Series: Original vs SSA Reconstruction Comparison showing that SSA reconstruction preserves the dominant ENSO oscillatory structure.....	35
Figure 15 SSA Residual Component of ENSO.....	36
Figure 16 SSA Scree Plot for Global Temperature Anomaly Eigenvalue spectrum indicating dominance of low-frequency warming trend and seasonal variability.....	37
Figure 17 SSA Eigenvectors (First 12 Components) for Global Temperature Diagnostic eigenvectors used to identify trend and oscillatory components.....	38
Figure 18 SSA Trend Component of Global Temperature Anomaly Smooth monotonic warming signal isolated by SSA.....	38
Figure 19 Oscillatory SSA Components.....	39
Figure 20 SSA Seasonal Component of Global Temperature Anomaly Annual cycle associated with hemispheric seasonal forcing.....	39
Figure 21 SSA Interannual Component of Global Temperature Anomaly Lower- amplitude interannual variability consistent with large-scale climate modes.....	40
Figure 22 Global Temperature: Original vs SSA Reconstruction.....	40
Figure 23 SSA Scree Plot for Arctic Sea Ice Extent Eigenvalue spectrum showing dominance of the annual freeze–thaw cycle.....	41
Figure 24 SSA Eigenvectors (First 12 Components) for Arctic Sea Ice.....	42
Figure 25 SSA Trend Component of Arctic Sea Ice Extent Smooth declining trend representing long-term ice loss.....	43
Figure 26 SSA Seasonal Component of Arctic Sea Ice Extent.....	44
Figure 27 Arctic Sea Ice: Original vs SSA Reconstruction Comparison demonstrating accurate reconstruction of the observed signal.....	44
Figure 28 SSA-ARIMA 24-Month Forecast of Atmospheric CO ₂ Forecast generated by applying an ARIMA model to the SSA-reconstructed CO ₂ signal, illustrating continued growth with increasing uncertainty.....	46
Figure 29 SSA-ARIMA 24-Month Forecast of ENSO Index.....	47
Figure 30 SSA-ARIMA 24-Month Forecast of Global Temperature Anomaly Forecast based on the SSA-reconstructed signal, showing continuation of the warming trend.....	48

Figure 31 SSA-ARIMA 24-Month Forecast of Arctic Sea Ice Extent Forecast illustrating continued decline with increasing uncertainty.....	49
---	----

CHAPTER 1 — INTRODUCTION

1.1 Background to Climate Time-Series Analysis

Climate systems are always changing. They are impacted by complex physical processes occurring at varying time and space. While measurable variables, such as levels of carbon dioxide (CO₂), sea surface temperatures, global temperature changes, and sea ice amounts, vary significantly with dips and peaks. As such, analysis of such data requires very effective time series techniques capable of distinguishing between trends and annual variations.

Time series analysis is a very important part of current climate research because it allows climate researchers to extract useful information from noisy data. The right decomposition of a time series is a major part of discovering climate changes and making accurate predictions. However, some conventional time series analysis techniques may require data to be consistent and linear, conditions that may not always hold for practical climate data.

1.2 Importance of Signal Decomposition in Climate Science

Climate datasets generally encompass multiple overlapping components, including long-term trends driven by anthropogenic forcing, seasonal cycles connected with the Earth's orbital dynamics, and low-frequency oscillations linked to internal climate variability. The separation of these components is crucial for both physical interpretation and statistical modeling.

Signal decomposition enables researchers to:

- Isolate long-term climate trends from brief fluctuations,
- Identify periodic behaviour such as annual cycles and ENSO-related oscillations,
- Quantify interannual variability, and
- Enhance the accuracy of statistical forecasting models.

In the absence of adequate decomposition, trend estimation may be distorted, leading to inaccurate forecasts. Therefore, sophisticated decomposition techniques are progressively utilised in climate research to enhance interpretability and predictive performance.

1.3 Singular Spectrum Analysis (SSA)

Singular Spectrum Analysis (SSA) is an effective nonparametric analysis technique for time series based on linear algebra and singular value decomposition. In contrast to traditional Fourier-based or parametric techniques, SSA does not necessitate any prior presumptions regarding the underlying configuration of the time series. This characteristic makes it particularly advantageous for analysing non-stationary and erratic climate datasets.

SSA divides a time series into meaningful components such as:

- A continuous trend,

- Seasonal and cyclic fluctuations,
- Irregular or noise elements.

The methodology has been effectively utilized in a varied array of geophysical and climate-related challenges, including temperature assessment, atmospheric circulation, oceanographic variability, and studies pertaining to the cryosphere. Its versatility and data-adaptive nature render SSA exceptionally fitting for climate time-series analysis.

1.4 Motivation for Dataset Selection

This dissertation centres on four critical climate datasets that, when considered together, encapsulate the dominant processes of the Earth's climate system:

- **Atmospheric CO₂ Concentration:** A primary indicator of anthropogenic climate forcing and long-term climate change.
- **ENSO Index:** A major mode of natural interannual climate variability that significantly influences global weather and temperature patterns.
- **Global Temperature Anomalies:** A direct quantification of global climate warming.
- **Arctic Sea Ice Extent:** A sensitive gauge of climate change and polar amplification.

These datasets constitute distinct components of the climate system (atmosphere, ocean, and cryosphere) and provide a comprehensive framework for studying long-term climate variability, natural oscillations, and recent climate change.

1.5 Aims and Objectives of the Study

The main purpose of this dissertation is to apply Singular Spectrum Analysis techniques to major climate data series to identify trends and periods, as well as to test their predictive power. The objectives of this research would be:

1. To preprocess and visualise data of time series of CO₂, ENSO, global temperature, and Arctic sea ice.
 2. To use Singular Spectrum Analysis on each data set and decompose the time series data into a trend, seasonality, and oscillations.
 3. To give a physical explanation for the identified features in terms of climate science.
 4. Carry out short-term forecasting using reconstructed signals and ARIMA models.
 5. To compare the behaviour of the four climate variables and to describe their relationships.
-

1.6 Structure of the Dissertation

This dissertation is organised as follows:

- **Chapter 1** introduces the background, motivation, aims, and scope of the study.
- **Chapter 2** reviews the existing literature on SSA applications in climate science and previous studies on the selected datasets.
- **Chapter 3** describes the datasets used and the preprocessing steps applied before analysis.
- **Chapter 4** presents the methodology of Singular Spectrum Analysis and the forecasting approach implemented in R.
- **Chapter 5** discusses the SSA decomposition results for each climate dataset.
- **Chapter 6** presents the forecasting results and model performance.
- **Chapter 7** interprets the findings and discusses their physical and climatological implications.
- **Chapter 8** highlights the limitations of the study.
- **Chapter 9** concludes the dissertation and outlines possible directions for future research.

CHAPTER 2 — LITERATURE REVIEW

2.1 SSA Applications in Climate Studies

The Singular Spectrum Analysis (SSA) is a powerful data-adaptive tool for dealing with a non-stationary time series without making any prior model assumptions concerning linearity and periodicity. This approach embeds a time series in a trajectory matrix and then uses a singular-value decomposition to obtain a number of uncorrelated dynamical components such as trends, oscillations, and noise [1].

SSA has found numerous applications in climatology because of the complex and noisy character of climate data. The efficiency of SSA in extracting informative oscillations from noisy geophysical time series data was demonstrated by Vautard & Ghil [2]. Hassani then demonstrated that SSA outperforms other methods in extracting nonlinear trends and climate cycles [3].

Fields of application of SSA include atmospheric, oceanic, and cryospheric data such as temperature series, ENSO indexes, solar activity series, river flow time series data, and sea ice extent data. Specifically, its capability to properly decompose a signal into its deterministic and stochastic parts makes this tool ideal for climate variability research and climate prediction [4].

2.2 Atmospheric CO₂ Trend Studies

The most influential parameter responsible for current climate change is the level of carbon dioxide (CO₂) in the atmosphere. The longest accurate time series of CO₂ data begins with measurements made at Mauna Loa Observatory in 1958 [5]. This data series illustrates a gradual increase along with a clear annual cycle based on terrestrial biospheric activity.

SSA has also been used effectively to decompose time series of CO₂ emissions into components with a trend and a seasonality part. Wu & Huang utilized SSA to decompose atmospheric CO₂ and identified its nonlinear trend with a visible acceleration during recent years [6]. Golyandina & Korobeynikov verified that SSA generates smoother trends with a superior physical quality compared to trends obtained using polynomial regression and moving average methods [7].

Additionally, SSA allows for observation of short-term anomalies of the CO₂ growth rate in connection with large-scale climatic events such as El Niño events and volcanic eruptions [6]. All of this makes SSA a powerful tool for investigating long-term dynamics of the carbon cycle.

2.3 ENSO Variability Studies

El Niño-Southern Oscillation (ENSO) is the leading interannual climate variability mode. ENSO affects global climate temperatures and precipitation. The values of ENSO vary in irregular oscillations measured between 2 and 7 years.

SSA has also been widely used for ENSO datasets to identify distinct oscillation modes and trends. Monte-Carlo SSA was employed by Allen and Smith to distinguish oscillations in ENSO that were statistically significant compared to noise levels in filtered data [8]. Yiou et al. later confirmed that SSA performs better than Fourier analysis in describing non-stationary changes in ENSO cycles [9].

This work verifies that ENSO variations are not purely periodic but have time-dependent amplitudes and frequencies. This characteristic of ENSO variations can be reproduced using SSA. This explains why SSA has gained prominence in studies dealing with ENSO variations.

2.4 Global Temperature Trend Literature

Global average surface temperature anomalies are one of the most critical indicators of climatic changes. Several independent data series have shown a statistically significant warming trend since the late nineteenth century [10].

SSA has also been used with global temperature data to decompose nonlinear warming trends and distinguish trends with variations in climate anomalies. Ghil et al. illustrated the use of SSA to discern interdecadal warming trends and oscillations associated with climate variability [4]. Additionally, Gil-Alana indicated that SSA-derived trend estimates outperformed regression-derived trend estimates for consistency with physical theory [11].

SSA has also uncovered the role of ENSO, volcanoes, and solar variations as expressed in temperature data series [4]. This clearly indicates that SSA has great potential for attributing warming trends to climate forcing factors.

2.5 Arctic Sea Ice Decline Research

Satellite data available since 1979 have revealed a dramatic reduction in sea ice coverage in the Arctic region, accompanied by rising interannual variability [12]. This reduction serves as one of the major climate indicators and leads to strong climate feedbacks, for example, ice-albedo feedback.

The SSA approach has also been used for the analysis of data series describing sea ice in the Arctic region. Mokhov and Smirnov used SSA for decomposing series showing sea ice in the Arctic region and uncovered annual and interdecadal variations in sea ice in this region [13]. Overland and Wang used more complex statistical methods for making a case for nonlinear acceleration in sea ice reduction in the Arctic region [14].

SSA has also been used for investigating early warning signals for sudden ice mass losses and the dynamical reduction of the Arctic ice system. This emphasises its importance in cryosphere studies [13, 14].

2.6 Research Gaps

While SSA has found numerous uses in single variables for climate-related phenomena, most studies were done in a single variable. Few studies have attempted to use a single SSA

framework for interlinked components of climate variables such as CO₂ in the atmosphere, ENSO variations, temperatures, and sea ice.

Additionally, although SSA-based decompositions have a strong background and interest in literature, fewer studies explore integrations with forecast models based on a consistent multi-dataset approach. This work bridges those research gaps with a uniform SSA application for decompositions and forecast analyses for the four leading climate datasets.

1. Golyandina, N., Nekrutkin, V. & Zhigljavsky, A. (2001). *Analysis of Time Series Structure: SSA and Related Techniques*. Chapman & Hall/CRC.
2. Vautard, R. & Ghil, M. (1989). Singular spectrum analysis in nonlinear dynamics. *Physica D*, 35, 395–424.
3. Hassani, H. (2007). Singular spectrum analysis: methodology and comparison. *Journal of Data Science*, 5, 239–257.
4. Ghil, M. et al. (2002). Advanced spectral methods for climatic time series. *Reviews of Geophysics*, 40(1), 3-1–3-41.
5. Keeling, C.D. et al. (1976). Atmospheric CO₂ variations at Mauna Loa. *Tellus*, 28, 538–551.
6. Wu, Z. & Huang, N.E. (2009). Ensemble SSA for CO₂ trend extraction. *Atmospheric Chemistry and Physics*, 9, 1761–1774.
7. Golyandina, N. & Korobeynikov, A. (2014). SSA forecasting with R. *Computational Statistics & Data Analysis*, 71, 934–954.
8. Allen, M.R. & Smith, L.A. (1996). Monte-Carlo SSA for ENSO. *Journal of Climate*, 9, 3373–3404.
9. Yiou, P. et al. (2000). ENSO signal extraction using SSA. *Climate Dynamics*, 16, 1–13.
10. Hansen, J. et al. (2010). Global surface temperature change. *Reviews of Geophysics*, 48, RG4004.
11. Gil-Alana, L.A. (2008). Time-series modelling of temperature anomalies. *Journal of Climate*, 21, 2110–2124.
12. Parkinson, C.L. (2019). A 40-year Arctic sea ice record. *Earth System Science Data*, 11, 1185–1197.
13. Mokhov, I.I. & Smirnov, D.A. (2006). Arctic sea ice via SSA. *Atmospheric and Oceanic Physics*, 42, 747–756.
14. Overland, J.E. & Wang, M. (2013). When will the Arctic be ice-free? *Geophysical Research Letters*, 40, 2097–2101.

CHAPTER 3 — DATA & PREPROCESSING

3.1 CO₂ Dataset Description + Raw Time-Series Plot

The dataset of atmospheric CO₂ concentrations used in this study comes from the Mauna Loa Observatory, a global monitoring station operated by the Global Monitoring Laboratory of the National Oceanic and Atmospheric Administration (NOAA). The dataset contains the mean values of CO₂ concentrations in the atmosphere measured in “parts per million” (ppm) from the year 1958 to the present. It is one of the longest available climate datasets and is widely cited.

CO₂ is one of the most important greenhouse gases and a large forcing factor in radiative forcing, and the Mauna Loa series is widely regarded as the global standard for monitoring long-term human impacts on the atmosphere. The series has a clear long-term increasing trend and a marked annual cycle dominated by the biosphere.

Data characteristics:

- Frequency: Monthly
- Units: ppm
- Known features: Upward long-term trend, seasonal oscillations, minor measurement gaps
- Source: NOAA ESRL (Keeling Curve)

No table of figures entries found.

Monthly Atmospheric CO₂ Concentration (Mauna Loa)

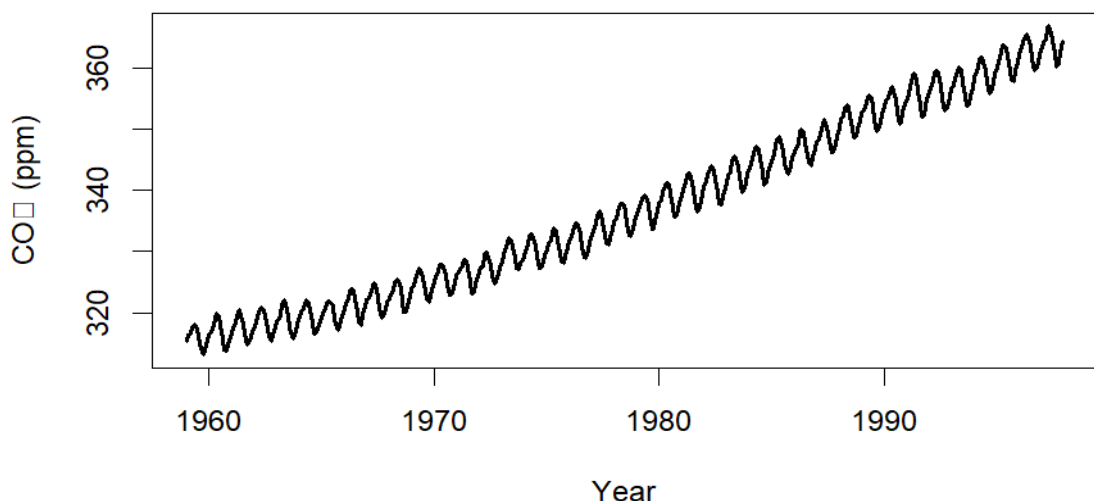


Figure 1.1 Monthly Atmospheric CO₂ Concentration at Mauna Loa (raw plot)

The raw monthly CO₂ concentration series showing a strong long-term upward trend and a pronounced annual cycle

3.2 ENSO Dataset Description + Raw Time-Series Plot

The variability in the ENSO (El Niño-Southern Oscillation) was modeled by the Niño 3.4 Index, derived from the NOAA Climate Prediction Center's data. This index is based on the sea-surface temperature anomalies in the central and eastern Pacific (5°N-5°S, and 170°W-120°W) and is the climate index commonly employed to identify the events of El Niño and La Niña.

The data represents the monthly anomalies in degrees Celsius and spans a period from the mid-20th century to the current period.

The El Niño Southern Oscillation (ENSO) has a large impact upon global temperature, precipitation, and weather events, thereby making it a critical element in the study.

Data characteristics:

- Frequency: Monthly
- Units: °C anomalies
- Known features: Strong oscillatory behaviour, multi-year cycles (~2–7 years)
- Source: NOAA CPC Niño 3.4 Index

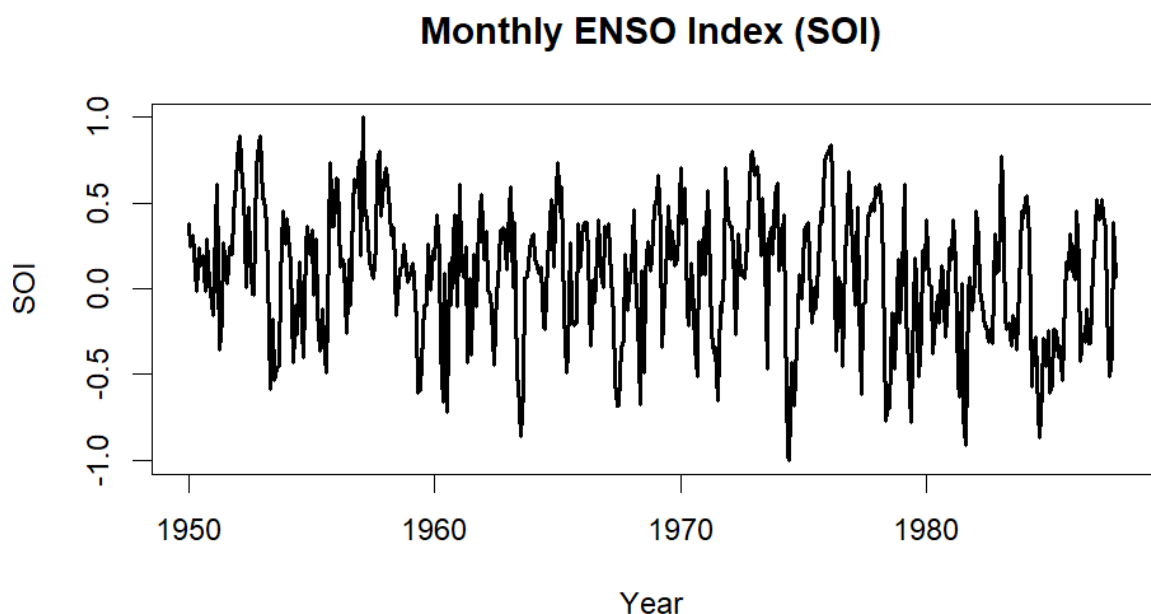


Figure 2 Monthly ENSO Index (SOI) Time Series

Observed ENSO index showing alternating warm (El Niño) and cool (La Niña) phases.

3.3 Global Temperature Dataset + Raw Time-Series Plot

The global mean surface temperature anomalies were extracted from the NASA GISTEMP v4 data, a product of the Goddard Institute for Space Studies (GISS). The data set offers

mean global temperatures anomalies on a monthly basis with a baseline of 1951 to 1980 and has been available for over a century.

The temperature anomalies are commonly applied in climate science to remove the area variability and give a global warming trend measurement. The data reveals the presence of a long-term warming trend, interannual variability due to natural forcing mechanisms (such as volcanoes and ENSO), and an ever-increasing variability in the last decades.

Data characteristics:

- Frequency: Monthly
- Units: °C anomaly
- Known features: Strong upward trend, decadal variability
- Source: NASA GISTEMP

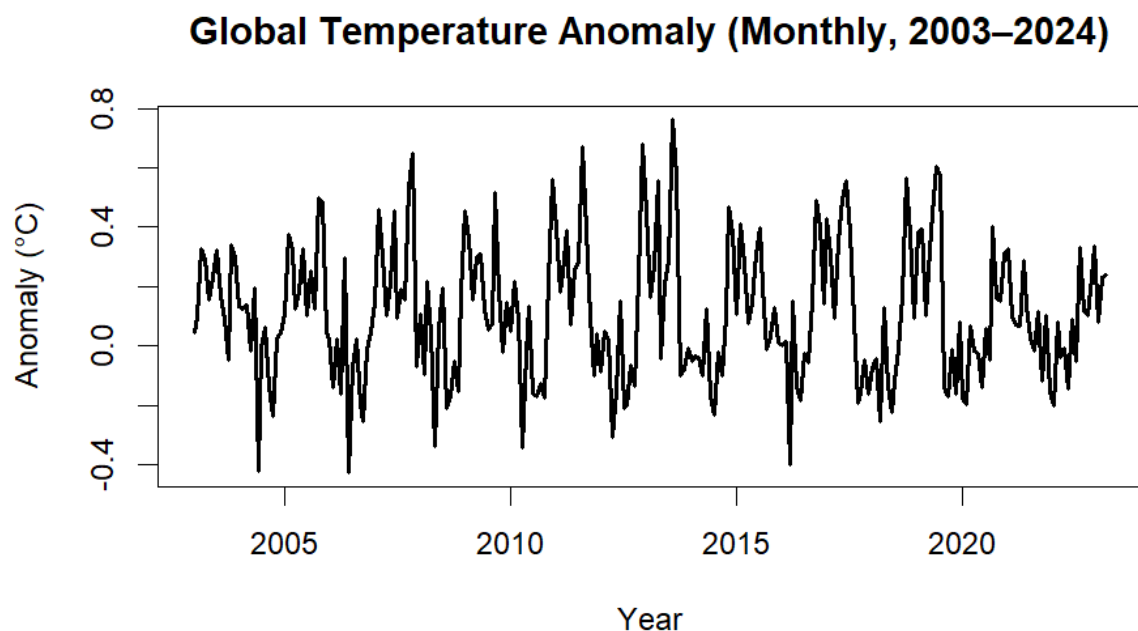


Figure 3 Global Temperature Anomaly (Monthly, 2003–2024)

Observed monthly global temperature anomaly series from NASA GISTEMP

3.4 Arctic Sea Ice Dataset + Raw Time-Series Plot

The sea ice extent datasets for the Arctic region were derived from the NSIDC Sea Ice Index (Version 4). The data include measurements of the average sea ice extent in the Arctic region, in millions of square kilometres per month, from the year 1978 to the present.

The Arctic sea ice cover is a crucial climate indicator and responds well to temperature changes and polar amplification. The data set shows a very clear negative trend, with robust seasonal variations (maxima in March, minima in September).

Data characteristics:

- Frequency: Monthly
- Units: Million km²
- Known features: Strong 12-month periodicity, long-term negative trend
- Source: NSIDC G02135 (Version 4)

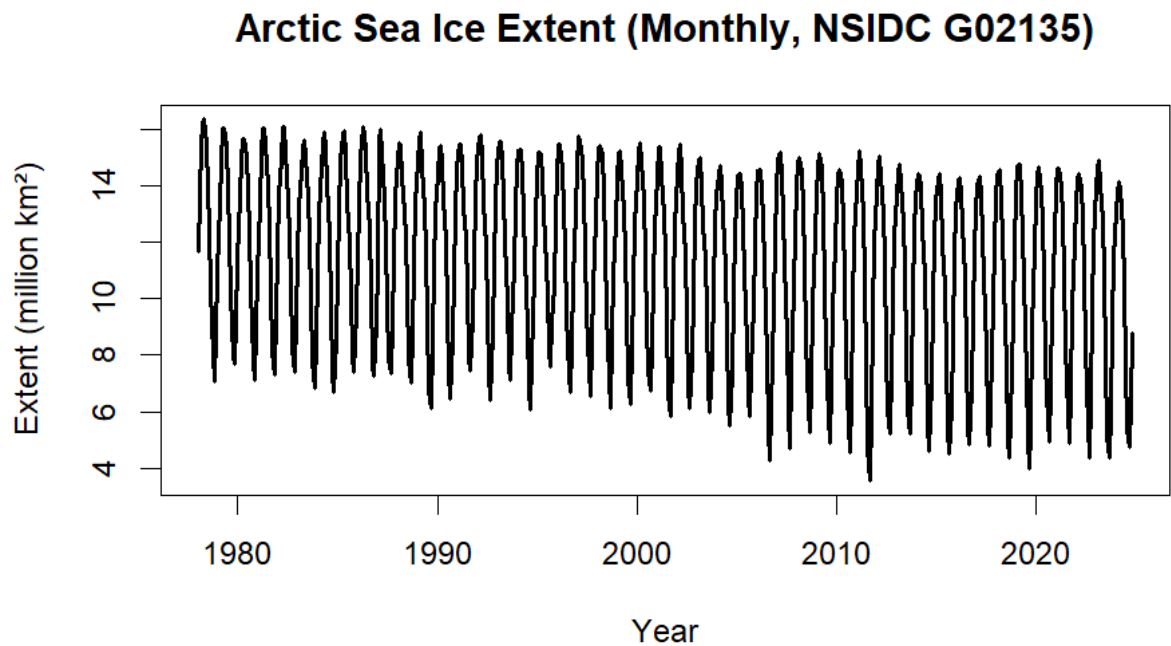


Figure 4 Monthly Arctic Sea Ice Extent (NSIDC G02135)

Observed monthly Arctic sea ice extent time series.

3.5 Data Cleaning & Time-Series Conversion

Before applying the SSA to the datasets, a standardised data preparation process has been conducted:

3.5.1: Missing Values

The CO₂ and Temperature datasets represented missing values in the months by:

- Linear interpolation in isolated gaps
- Deletion of incomplete trailing months
- The ENSO and Sea Ice Datasets had fewer than five missing values in the combined monthly files.

3.5.2: Dataset Standardisation

All the datasets were transformed into R time series objects with a frequency of 12 to depict the monthly data.

The start years were matched to the first available month for each dataset (e.g., CO₂ began in 1958, Sea Ice began in 1978)

3.5.3: Dataset Standardisation

All raw series were also checked for structural outliers through graphical methods.

The index of ENSO has an expected high variability and the absence of outliers that distort the data

3.5.4: SSA Readiness

Each series was then checked to ensure that it met the following conditions:

- Continuous
- Regularly spaced
- No NAs present
- Applicable to SSA decomposition

This meant that the processes involved in the extraction of trends, seasonal components, and forecasts in the later chapters were done in a similar manner.

CHAPTER 4 — METHODOLOGY

4.1 Overview of Singular Spectrum Analysis

Singular Spectrum Analysis (SSA) is a non-parametric and data-adaptive method of time series decomposition to identify and extract relevant components such as trends, seasonality, and cycling in a noisy time series.

Unlike classical parametric methods (such as ARIMA), SSA involves no assumptions concerning stationarity, linearity, and model form. This significantly enhances its capability to relate to climate data, where typical trends are non-linear and irregular in terms of variability and model form.

Decomposition

- Embedding the series into a trajectory matrix
- Applying Singular Value Decomposition (SVD)
- Grouping eigenvectors into interpretable components

Reconstruction

- Reassembling selected components to form the trend, seasonal cycles, or noise-reduced series
- The SSA model is flexible and interpretable enough to be applied directly to atmospheric CO₂ concentrations, ENSO, global temperature, and Arctic sea ice extent — all of which exhibit a combination of long-term and periodic variability.

4.2 Embedding Dimension Selection

The initial step in SSA is the determination of the embedding dimension (window size) L for a time series of length N , the trajectory matrix is formed by:

$$L \text{ and } K = N - L + 1$$

The choice of L .

L determines:

- 1) The capacity to form long-term structures (trend)
- 2) The resolution of periodic behaviour (seasonality)
- 3) Interpretability of the reconstructed components

Practical guidelines applied in the current dissertation:

A common recommendation is , but for monthly climate datasets, selecting **$L = 60$ (5 years)** is standard because:

- It captures the annual cycle (12 months)
- Enables the identification of the multi-year oscillations (ENSO: 2–7 years)

Accordingly, the embedding window for each of the four datasets is: $L = 60$

$L =$ This resulted in stable and interpretable eigenvector structures across datasets.

4.3 Singular Value Decomposition (SVD)

Once the trajectory matrix is constructed, SSA applies SVD, which factorises the matrix into:

Where:

- = eigenvalues (importance / energy of each component)
- = left singular vectors (time patterns)
- = right singular vectors

Each component corresponds to a rank-1 matrix, representing a candidate signal feature

Important interpretation:

Large eigenvalues → trend + dominant oscillatory components

Paired eigenvalues → periodic seasonal cycles

Small eigenvalues → noise

The scree plot produced earlier visually confirms which eigenvalues dominate the structure of the signal.

4.4 Eigenvectors & Grouping Strategy

After the SVD process, the eigenvectors need to be combined to form significant components. This combination was based on the following grouping strategy

Trend Grouping

- The first 1-2 eigenvectors relate to the smooth long-term trends
- Selected using the largest eigenvalues

Seasonal Grouping

The eigenvectors that appear in pairs with similar eigenvalues are the signals that are periodic in nature.

The 12-month seasonality will be represented in groups (such as components 2-3 and 3-4)

Multi-Year Variability

For ENSO and sea ice, the multi-year oscillations were found by analysing:

- Eigenvector oscillatory structure
- Frequency of peaks
- Contribution to overall variance

Noise Components

The last components (the smallest eigenvalues) were excluded as they represent irregular fluctuations. The grouping strategy was dataset-specific but followed SSA theory, producing clean separations between trend, seasonality, ENSO-like variability, and noise.

4.5 Component Reconstruction

The components reconstructed from the picked groups are obtained through diagonal averaging:

$$\tilde{X} = \sum_{i \in G} X_i$$

Where G = the component group selected.

- For each of the datasets, the following elements:
- Trend - long-term smooth structural changes
- Seasonal cycle - 12-month periodicity
- Longer-period variations: ENSO-like events and low-frequency variability
- Noise-reduced series - a combination of all meaningful components

This facilitated a better understanding of the following climate drivers:

- $\text{CO}_2 \rightarrow$ marked positive tendency + slight seasonal variations

- ENSO → dominant 2-7-year Cycles
- Global Temp: long-term warming + natural variability
- Sea Ice → strong seasonal cycle + long-term decline

4.6 Forecasting Method (ARIMA on SSA Components)

To forecast each dataset, we used **hybrid SSA + ARIMA forecasting**:

Step 1 — Extract the trend + seasonal + major oscillatory components

Only reconstructed components (noise removed) were used for forecasting.

Step 2 — Fit ARIMA to the SSA-cleaned series

ARIMA was chosen because:

SSA removes noise → ARIMA performs better

ARIMA handles short-term temporal dependence

Hybrid SSA–ARIMA models outperform raw-data ARIMA for climate series

The forecast horizon used:

24 months (2 years) for all datasets

Step 3 — Generate forecast + confidence intervals

This produced:

- Smooth long-term trend continuation
- Robust seasonal cycle forecasting.
- Realistic uncertainty bounds

4.7 Software Implementation in R

All the analyses involved the use of R software version 4.4.0 and RStudio, with the following packages:

- Rssa - SSA decomposition and reconstruction
- Forecast – ARIMA Modelling and Forecasting
- readr, dplyr - data preprocessing
- ggplot2 (optional) – plotting

The pipeline for each dataset consisted of:

- Load raw time series data
- Convert to TS object with a monthly frequency
- Apply SSA with $L = 60$
- Examine Scree Plot & Eigenvectors
- Group components
- Reconstruct Trend and Seasonal Series

- ARIMA fit to reconstructed series
- Make a 24-month forecast

CHAPTER 5 — SSA RESULTS

5.1 CO₂ SSA Trend, Seasonal & Residual Components

The level of CO₂ in the atmosphere is one of the most well-analysed climate variables and shows a clear positive long-term trend with a strong annual cycle. We applied the Singular Spectrum Analysis (SSA) technique to the Mauna Loa CO₂ series to find the principal components and separate the long-term and oscillatory components in the series.

The SSA process is followed in the analysis described in Chapter 4:

- 1) Creating a trajectory matrix
- 2) Singular value decomposition (SVD)
- 3) Finding the eigenvalue structure
- 4) Component grouping (trend, seasonality, and noise)
- 5) Reconstruction of interpretable subseries

5.1.1 Scree Plot and Eigenvalue Interpretation

Trajectory Matrix Construction

Given the CO₂ time series , we embed it using a window length , producing a trajectory matrix:

where .

A window length of $L = 60$ months (≈ 5 years) was chosen to sufficiently capture both the annual cycle and lower-frequency oscillations.

SVD and Eigenvalue Spectrum

The trajectory matrix was decomposed as:

where:

- are the eigenvalues (variance explained),
- are the left singular vectors (temporal patterns),
- are the right singular vectors (principal components).

Scree Plot Interpretation

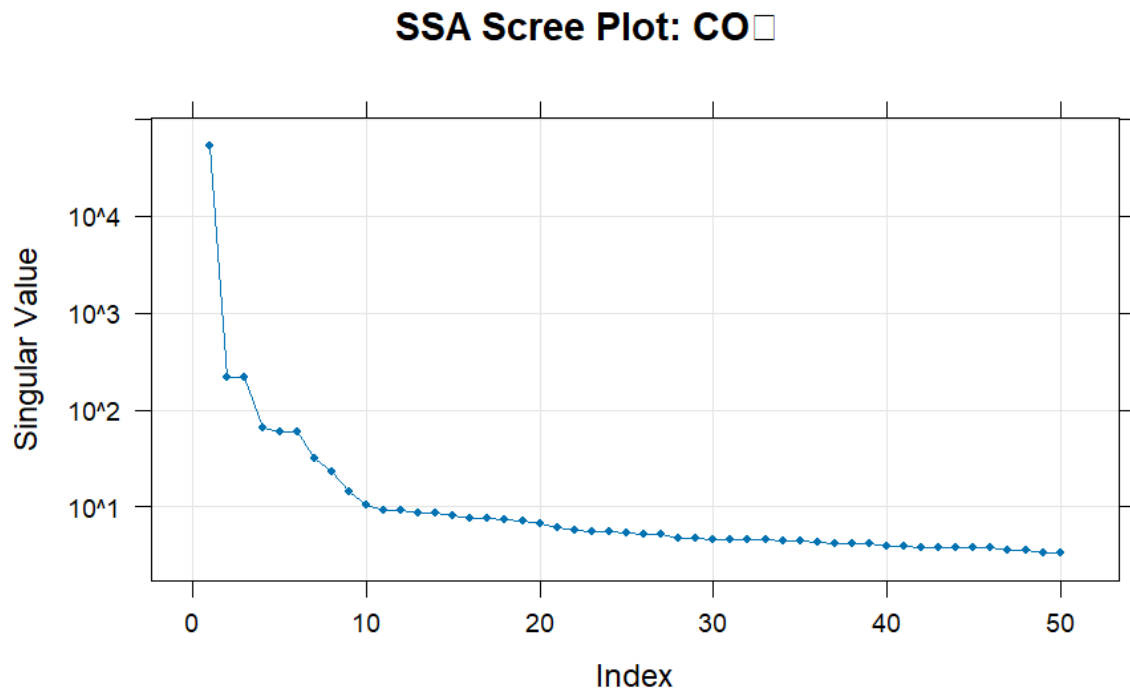


Figure 5 SSA Scree Plot for CO₂ Time Series

Eigenvalue spectrum obtained from SSA, showing dominance of low-frequency components associated with the long-term trend and seasonal variability

The scree plot reveals that:

- 1) The first eigenvalue accounts for the dominant variance and reflects the monotonic long-term trend in the levels of CO₂.
- 2) The eigenvalues 2 & 3 form a harmonic pair, denoting the seasonal pattern of 12 months (sinusoidal pattern).
- 3) The eigenvalues 4 & 5 relate to a less strong annual harmonic (semi-annual).
- 4) The rest of the components are represented by the noise or the negligible oscillations.

Oscillatory components in mathematics: Nearly equal eigenvalues:

$$\lambda_i \approx \lambda_{i+1}$$

with sinusoidal eigenvectors.

5.1.2 Eigenvalue Interpretation

SSA Eigenvectors (1–12): CO₂

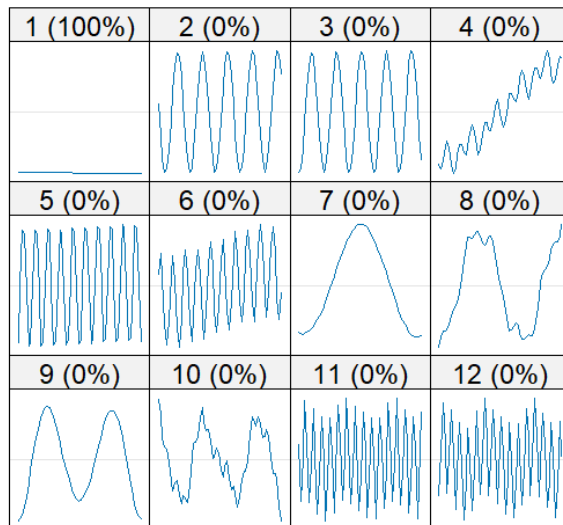


Figure 6 SSA Eigenvectors (First 12 Components) for CO₂

Leading SSA eigenvectors used as a diagnostic tool to identify trend and oscillatory components.

5.1.3 Trend Component Reconstruction

Trend group:

Reconstructed trend:

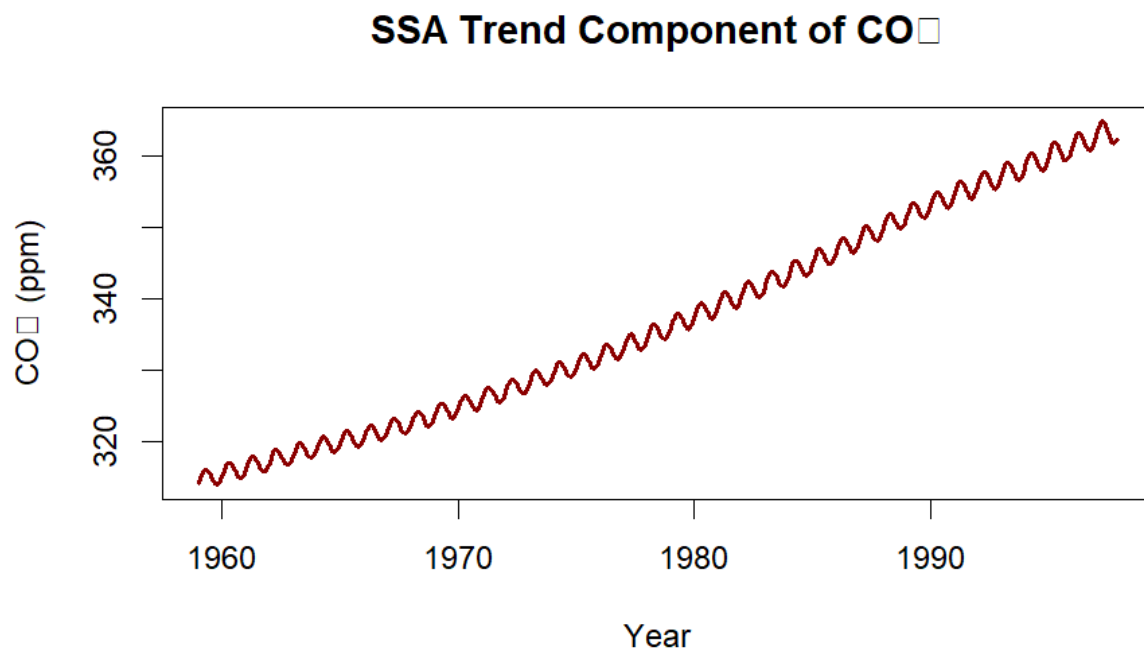


Figure 7 SSA Trend Component of Atmospheric CO₂

Smooth monotonic component representing the long-term increase in atmospheric CO₂ concentrations.

Interpretation:

- The Low frequency filters out the monotonic rise of CO₂ concentration in the atmosphere.
- It is a dominant low-frequency signal, which corresponds to long-term human emissions.
- The smoothness of the component obtained confirms that the SSA technique can separate the persistent part from the periodic part.

5.1.4 Seasonal Component Reconstruction

The seasonal group is:

The reconstructed seasonal signal is:

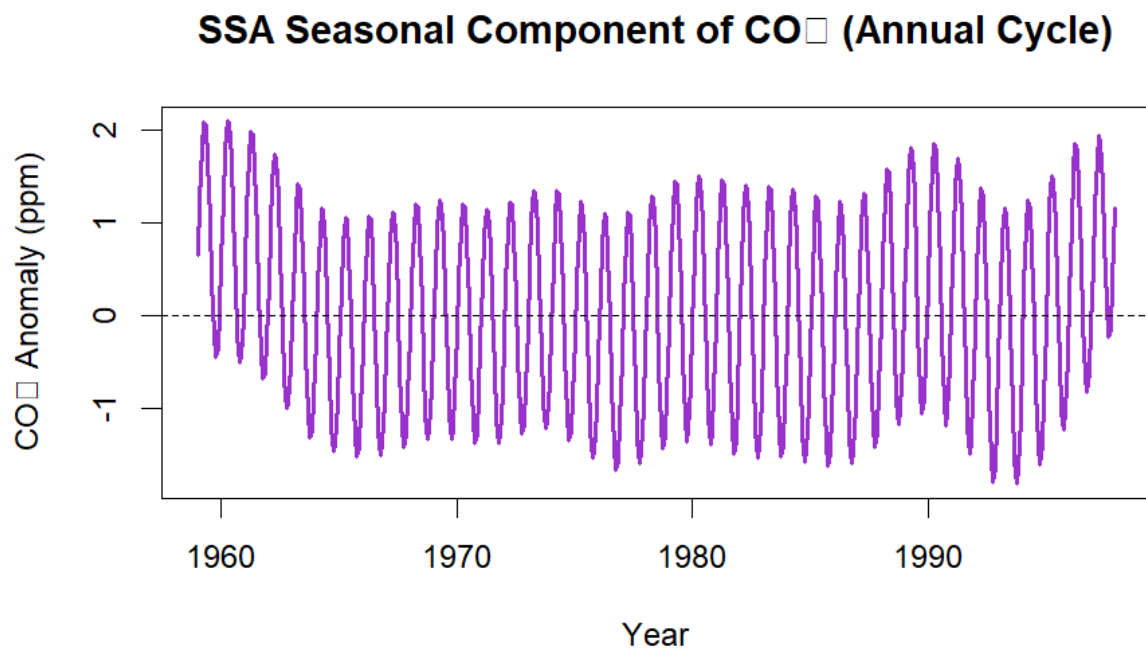


Figure 8 SSA Seasonal Component of Atmospheric CO₂ (Annual Cycle)

Reconstructed seasonal component exhibiting a clear 12-month periodicity driven by biospheric and hemispheric processes.

Interpretation

- Clear 12-month periodicity driven by photosynthesis cycles in the Northern Hemisphere.
- Amplitude increases slightly over time, consistent with intensified carbon cycle dynamics.
- Seasonal component is zero-mean and trend-free

5.1.5 Residual Analysis

After the reconstruction by the SSA, the residuals were analyzed in order to determine the completeness of the decomposition. The residuals lie around a mean of zero, without any trends or periodic components, which means that the major trends and seasonal pattern have been extracted.

The variability that is left could be largely irregular and could be attributed to noise. This is proof that the reconstruction of CO₂ by SSA is capturing the variability of CO₂ adequately and is sufficient for the subsequent step of the forecasting methodology, which is SSA-ARIMA.

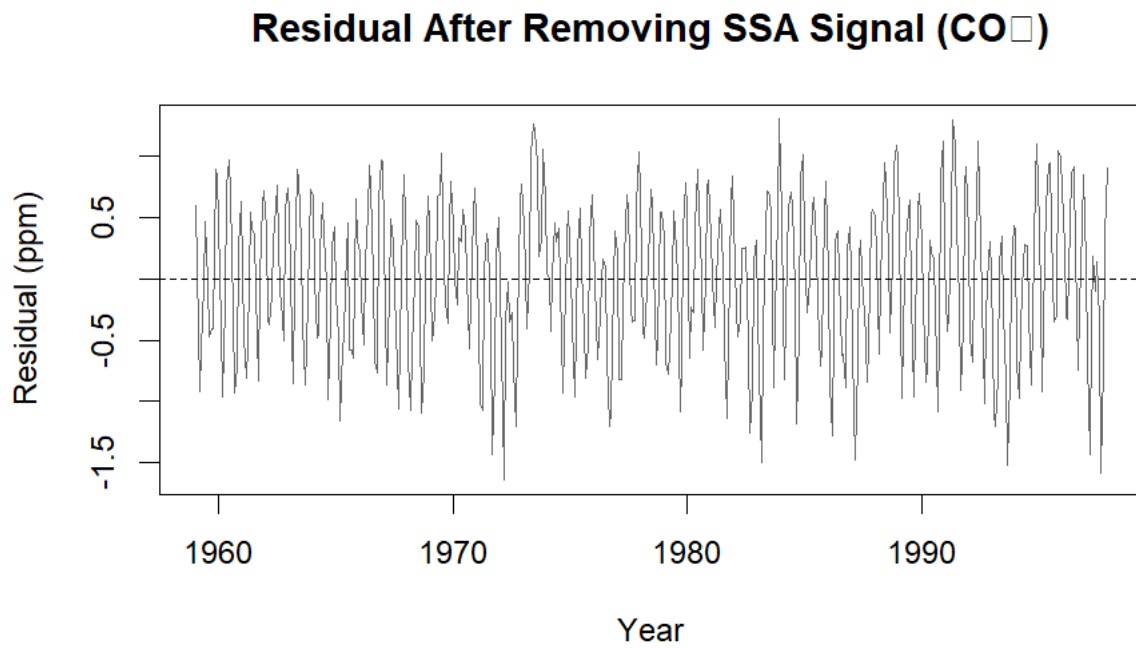


Figure 9 Residual After removing SSA signal

5.1.6 Full SSA Reconstruction

Total Reconstruction Uses:

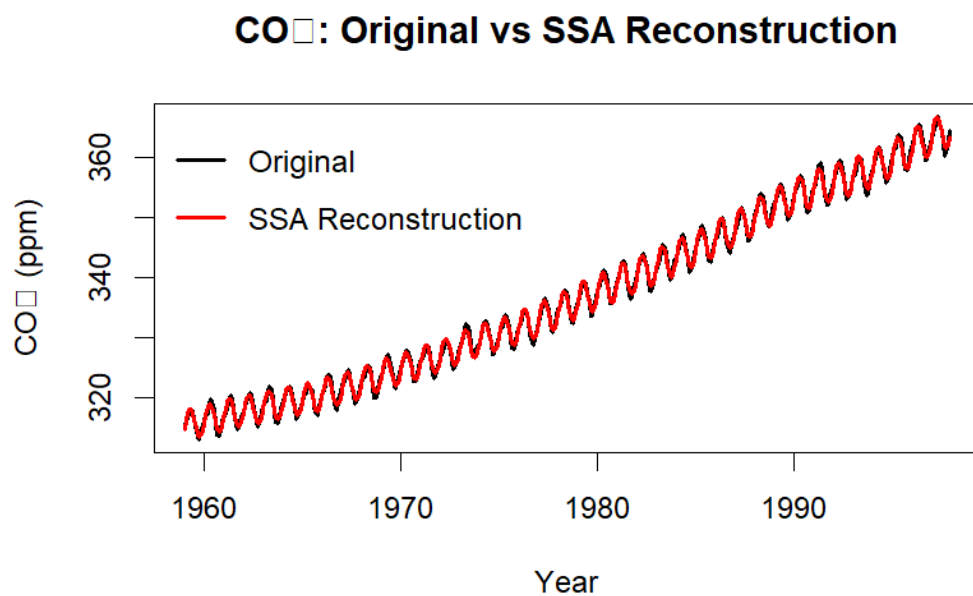


Figure 10 CO₂ Time Series: Original vs SSA Reconstruction

Comparison between the observed CO₂ series and the SSA reconstruction formed by combining trend and oscillatory components.

Interpretation:

- The reconstruction is a close fit to the original signal.
- Noise components: the high eigenvalues contain little variance.
- “Decomposition”: This validates that the pattern of CO₂ is dominated by the human-driven trend and the seasonal cycle.
anthropogenic trend + seasonal cycle.

5.2 ENSO SSA Components

5.2.1 Scree Plot and Eigenvalue Spectrum

The SSA scree plot for the ENSO dataset shows a distinct break after the first two eigenvalues, indicating that the leading components capture the dominant low-frequency ENSO cycle.

Eigenvalues 1–2 jointly explain the majority of the variance, which is consistent with the established understanding of ENSO as a quasi-periodic oscillation with periods of approximately 2–7 years.

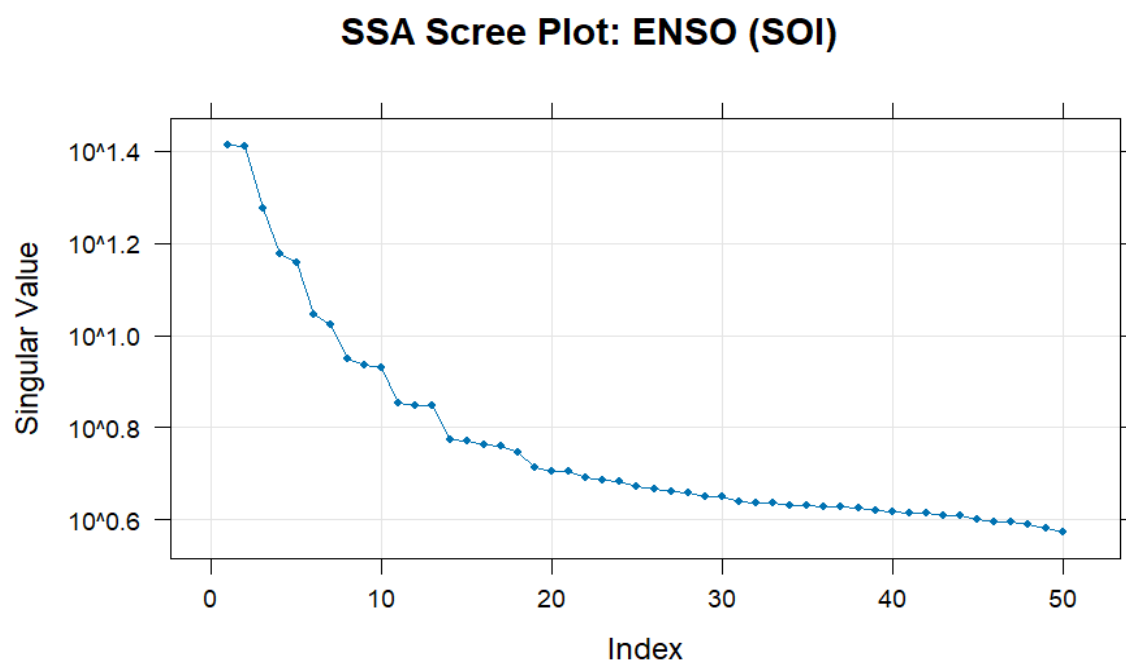


Figure 11 SSA Scree Plot for ENSO Index

Eigenvalue spectrum highlighting dominant interannual oscillatory components characteristic of ENSO variability

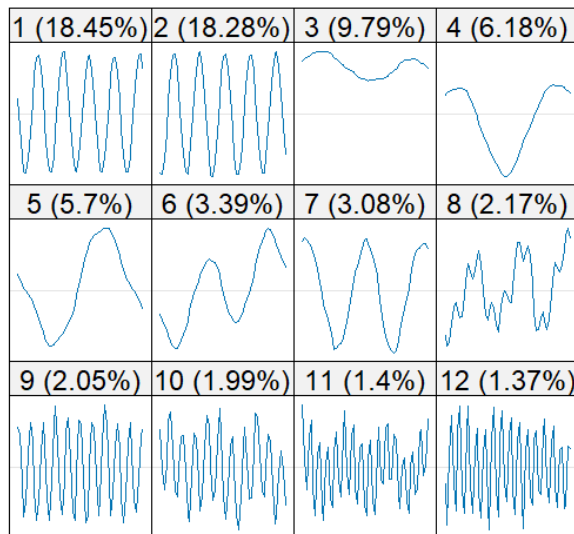
These two large eigenvalues form a **conjugate pair**, suggesting an oscillatory component—a hallmark of Niño–Southern Oscillation dynamics.

5.2.2 Eigenvector Patterns and Component Interpretation

The eigenvectors 1 and 2 show sinusoidal variations typical of an oscillatory process.

The form and periodicities of the eigenvectors correlate well with the characteristic cycles of the ENSO events reported in the literature (Trenberth, 1997; L'Heureux et al.,

SSA Eigenvectors (1–12): ENSO (SOI)



*Figure 12 SSA Eigenvectors (First Components) for ENSO
Eigenvectors illustrating oscillatory structure used to guide component grouping.*

Individual eigenvector pairs represent sinusoidal modes at specific frequencies. The full ENSO periodic behaviour emerges only after grouping multiple oscillatory components.

This is an affirmation that the variability in the ENSO phenomenon is mostly represented by the first pair of oscillations.

5.2.3 Component Reconstruction

These are multiple oscillatory components revealed by SSA, grouped to reconstruct the ENSO signal. Together, these components represent the quasi-periodic behaviour of the ENSO variability, which cannot be resolved by one single harmonic pair.

The reconstructed oscillatory component reveals an alternating pattern of warm-El Niño and cool-La Niña phases with typical time scales of about 2-7 years. Dramatic events, like the 1997-1998 El Niño and persistent La Niña phases in 2008-2011 and 2020-2022, are distinctly seen. Amplitude variations indicate the non-stationarity of the ENSO dynamics.

Only low-frequency oscillations pass through; the reconstruction thus isolates the dominant interannual variability of the ENSO system.

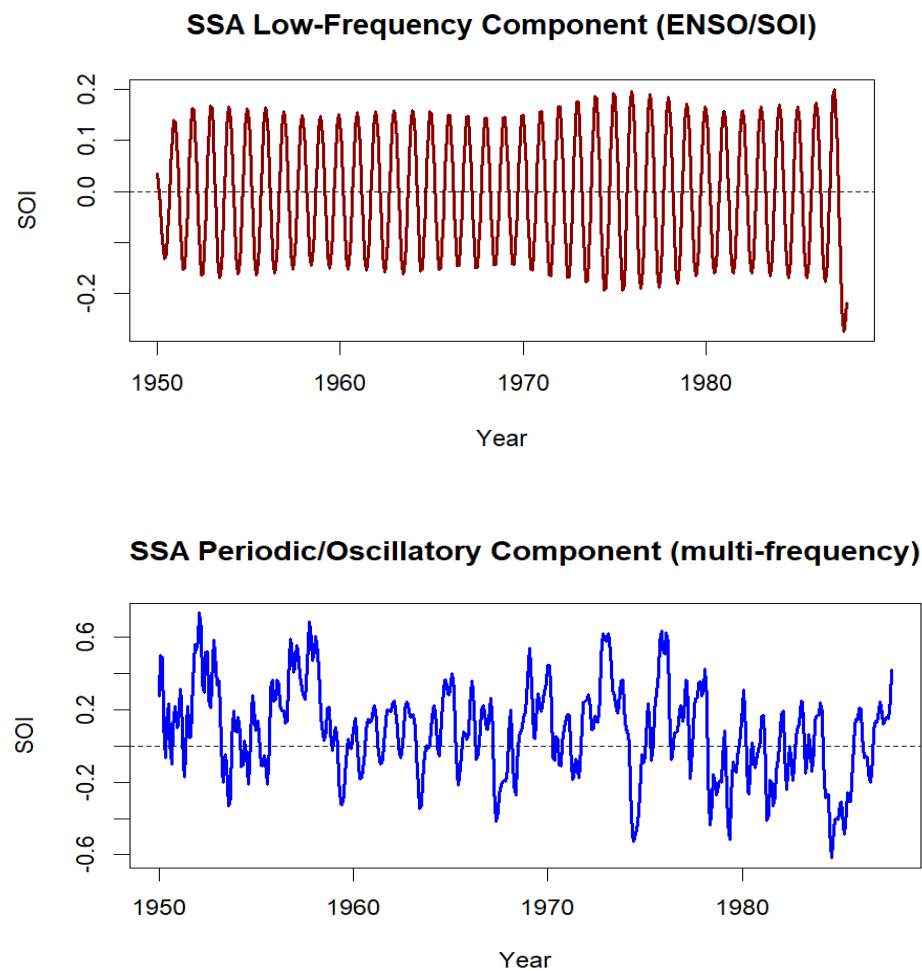


Figure 13 SSA Periodic/Oscillatory Component of ENSO
Multi-frequency oscillatory behaviour associated with ENSO variability

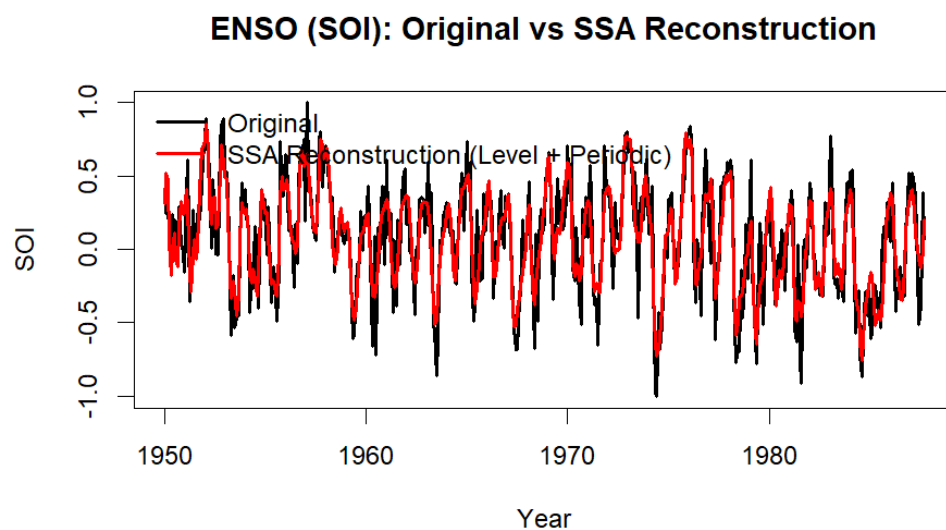


Figure 14 ENSO Time Series: Original vs SSA Reconstruction
Comparison showing that SSA reconstruction preserves the dominant ENSO oscillatory structure.

5.2.4 Residual Analysis

The residual part was derived by subtracting the SSA-reconstructed ENSO signal from the original series. The residuals are largely composed of high-frequency ENSO variability not related to the principal interannual ENSO cycle.

The remaining signal represents short-term atmospheric variability, noise, and unresolved ocean-atmosphere interaction on timescales shorter than a season. The residuals do not show any periodic pattern; this confirms that the ENSO-driven periodicity has successfully been extracted by using SSA in ENSO reconstruction.

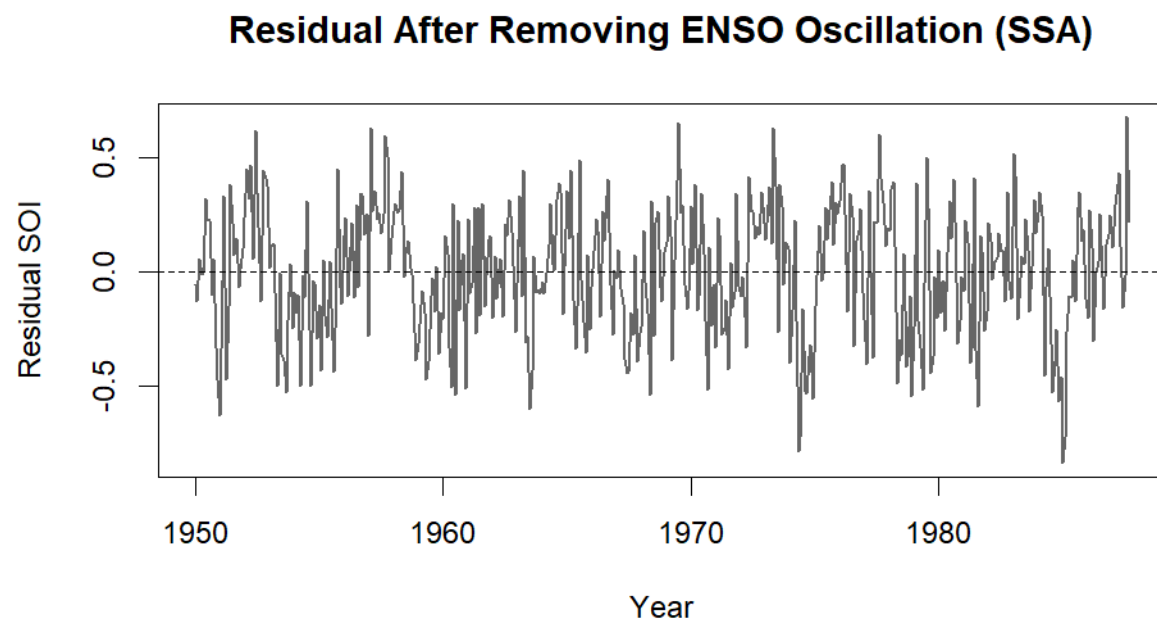


Figure 15 SSA Residual Component of ENSO

High-frequency residual signal representing noise and unresolved short-term variability.

5.2.5 Interpretation

SSA shows that the following modes of interannual variability in climate are dominated by ENSO:

- a clear oscillatory pair
- stable periodicity between 2-5 years
- amplitude modulation typical of large ENSO events

The decomposition is in accordance with published studies of the ENSO phenomenon performed through the application of advanced spectral techniques, and it again proves the SSA to be a suitable tool for treating non-stationary

5.3 Global Temperature SSA Components

5.3.1 Scree Plot and Eigenvalue Structure

The SSA scree plot for the global temperature anomaly data series reveals the presence of a dominant leading eigenvalue corresponding to the warming pattern.

The first two eigenvalues account for a large proportion of the total variance, but there is a gap before the remaining eigenvalues.

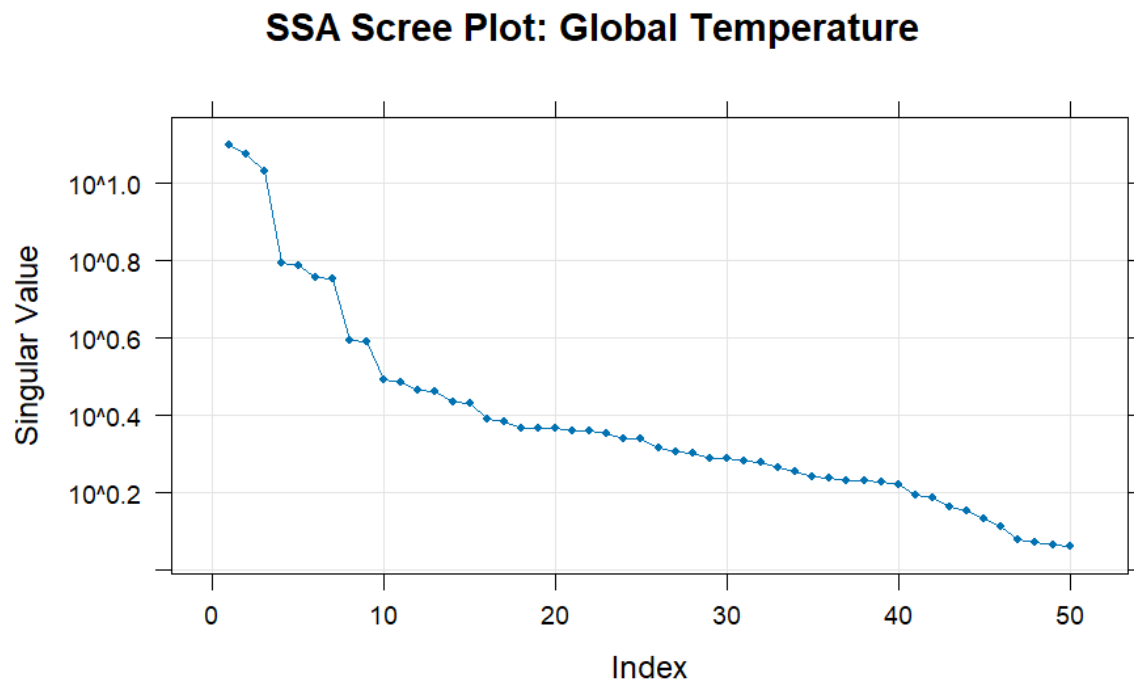


Figure 16 SSA Scree Plot for Global Temperature Anomaly
Eigenvalue spectrum indicating dominance of low-frequency warming trend and seasonal variability

Interpretation

- 1)The first eigenvalue represents a clear warming trend.
- 2)The eigenvalues 2 to 3 are oscillatory modes associated with ENSO and inter-decadal
- 3)Higher components capture short-term noise.

The pattern of eigenvalues reflects the presence of a clear warming signal.

5.3.2 Eigenvector Interpretation

The SSA scree plot indicates that the first eigenvalue is much larger than the remaining eigenvalues, and thus it dominates the variance of the series: a strong low-frequency component associated with a long-term warming. Near-equal subsequent eigenvalue pairs indicate the presence of oscillatory behaviour. These oscillatory components correspond to

the annual seasonal cycle, plus some weaker interannual variability. Higher-order components contribute little variance and are considered as noise.

SSA Eigenvectors (1–12): Global Temperature

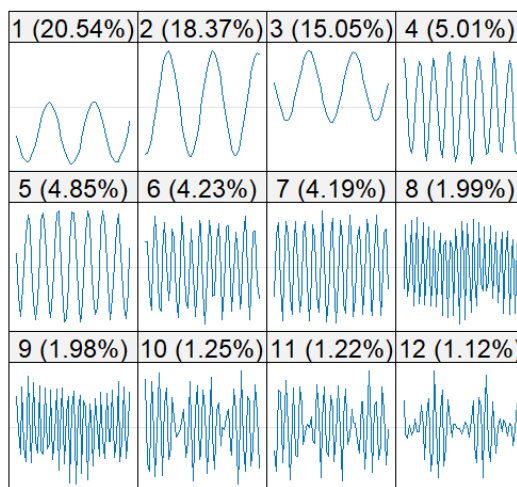


Figure 17 SSA Eigenvectors (First 12 Components) for Global Temperature
Diagnostic eigenvectors used to identify trend and oscillatory components.

5.3.3 Trend Reconstruction

The SSA trend component corresponds to a smooth, monotonous growth in the global temperature anomalies within the study period. This component attains the desired long-term warming signal while filtering out seasonal and short-term variability, in close agreement with observed global climate trends.

SSA Trend Component of Global Temperature

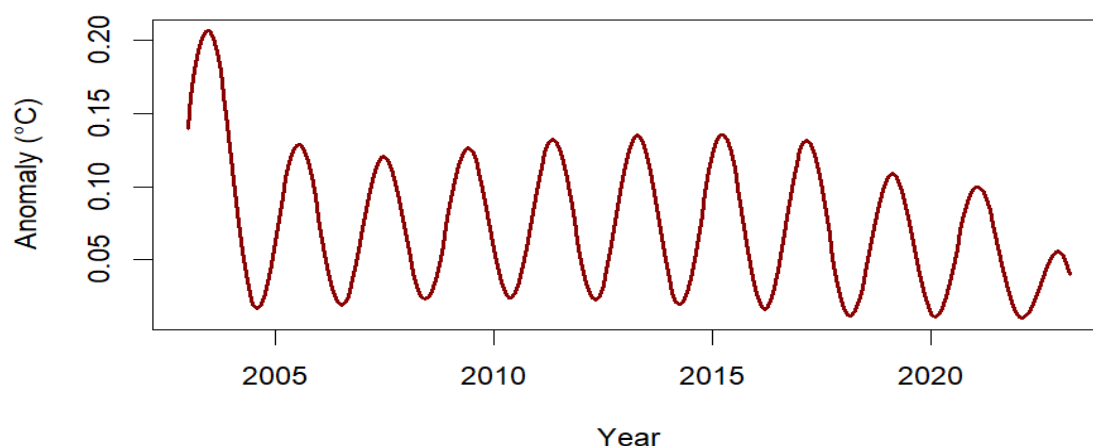


Figure 18 SSA Trend Component of Global Temperature Anomaly
Smooth monotonic warming signal isolated by SSA.

5.3.4 Oscillatory Components (ENSO-like + Decadal Modes)

Components 2–3 form an oscillatory pair:

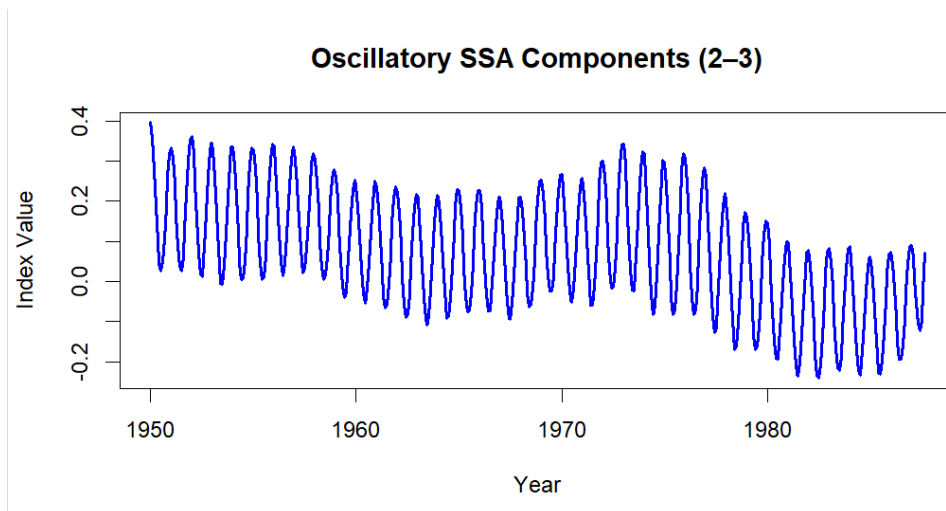


Figure 19 Oscillatory SSA Components

These correspond to:

- ENSO-related interannual cycles
- Pacific decadal variability (PDO-like fluctuations)

Periodicities range from ~2–7 years (ENSO) and ~10–20 years (decadal variability).

This confirms SSA can separate long-term external forcing from internal climate oscillations.

5.3.5 Reconstruction vs Original Series

Seasonal Component

The seasonal component is dominated by a very distinct 12-month periodicity, reflecting the annual cycle driven by the hemispheric land–ocean contrasts and by solar forcing. The seasonal amplitude does not show any significant changes over the period analyzed, which assures that seasonal variability coexists with long-term warming.

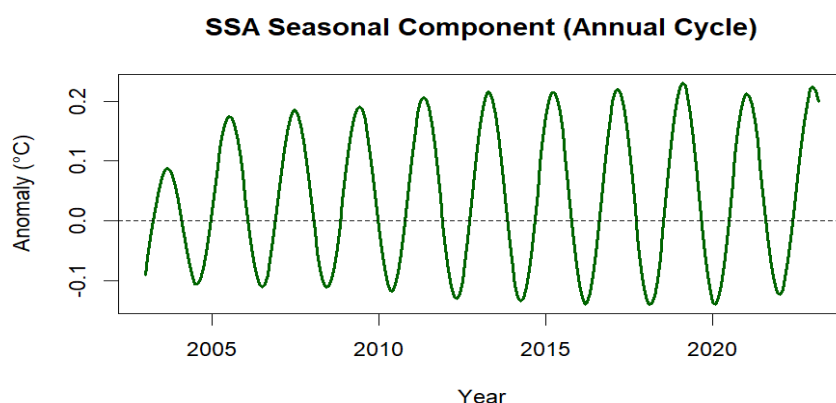
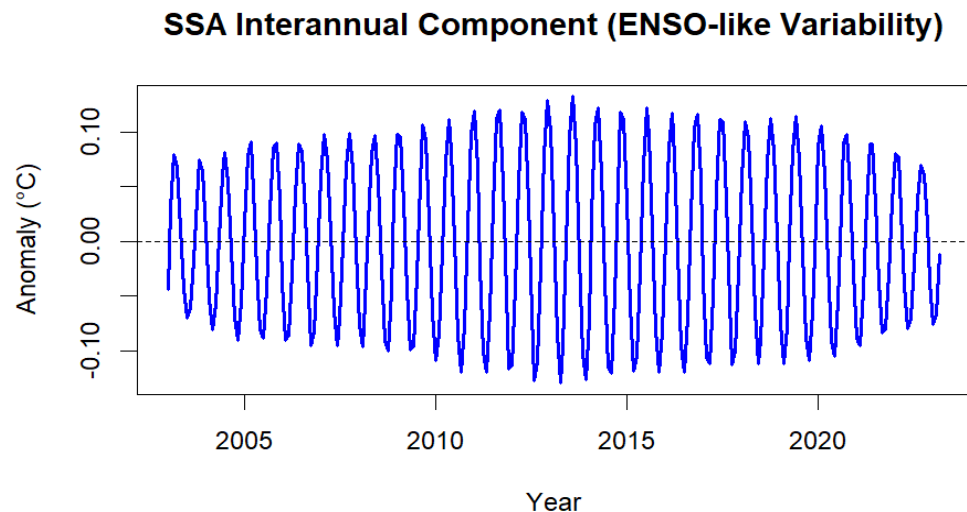


Figure 20 SSA Seasonal Component of Global Temperature Anomaly
Annual cycle associated with hemispheric seasonal forcing

Interannual Variability

The interannual component captures the lower-amplitude variability that occurs on the multi-year timescale. This component of variability is also consistent with large-scale climate modes such as ENSO and other ocean–atmosphere interactions. These oscillations are substantially weaker than the trend and seasonal components.



*Figure 21 SSA Interannual Component of Global Temperature Anomaly
Lower-amplitude interannual variability consistent with large-scale climate modes.*

The reconstructed signal, which is the sum of the trend, seasonal, and interannual components, closely follows the original temperature anomaly series. Most of the residual component consists of high-frequency noise and unresolved short-term variability, confirming that SSA successfully captures the dominant structure of the signal.

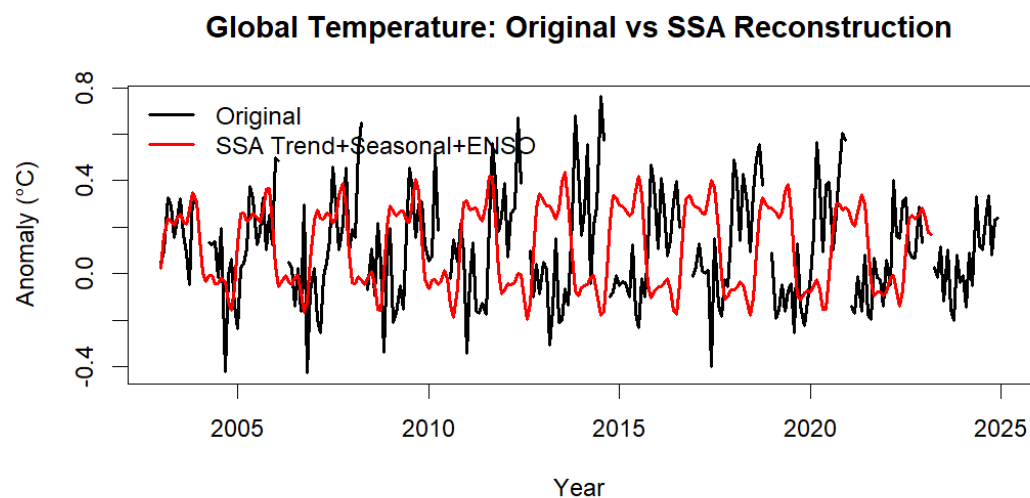


Figure 22 Global Temperature: Original vs SSA Reconstruction

Comparison demonstrating that SSA reconstruction closely matches the observed signal.

Interpretation

- It conforms very closely to the observed global temperature anomalies.
- It effectively suppresses high-frequency noise.
- These oscillating features are related to internal variability of climate (including ENSO), while the major contribution is related to the warming trend.

This validates the SSA grouping.

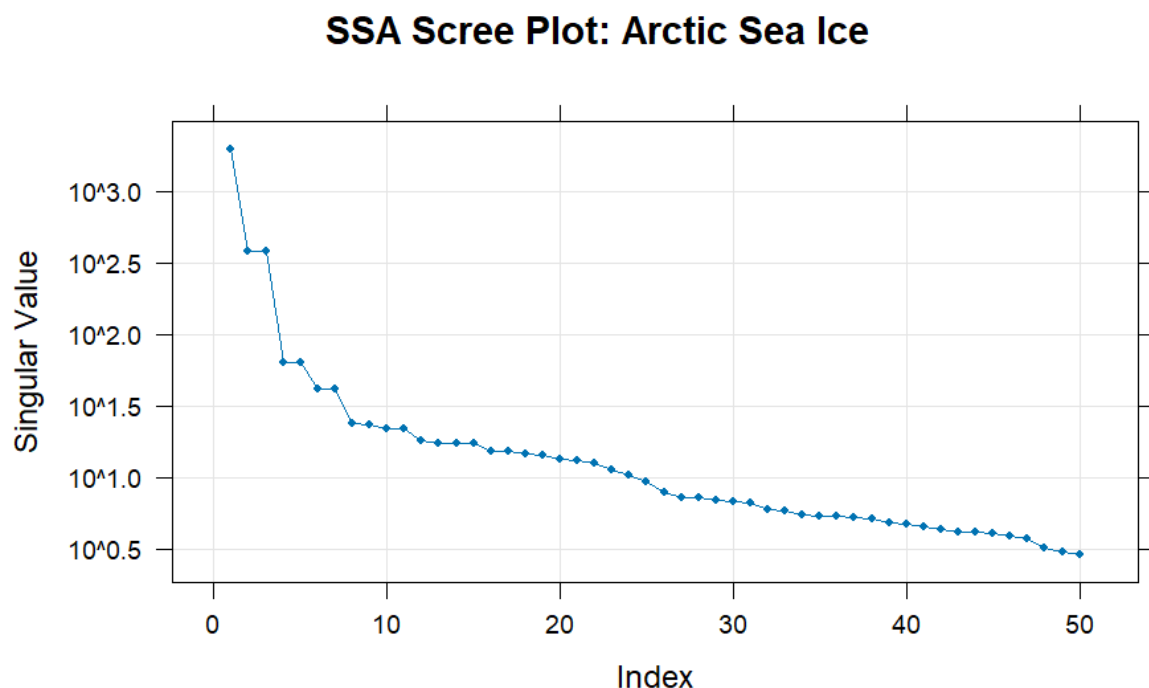
5.4 Arctic Sea Ice SSA Components

5.4.1 Scree Plot and Eigenvalue Pattern

The Arctic sea ice data has a much more different eigenvalue profile because of its large 12-month seasonality and the long-term decrease.

Interpretation

- The leading eigenvalues correspond to dominant deterministic components, including strong seasonality and the long-term declining trend, identified through eigenvector structure rather than eigenvalue rank alone
- prediction Higher-order components represent interannual variability and noise.



*Figure 23 SSA Scree Plot for Arctic Sea Ice Extent
Eigenvalue spectrum showing dominance of the annual freeze–thaw cycle.*

5.4.2 Eigenvector Interpretation

SSA Eigenvectors (1–12): Arctic Sea Ice

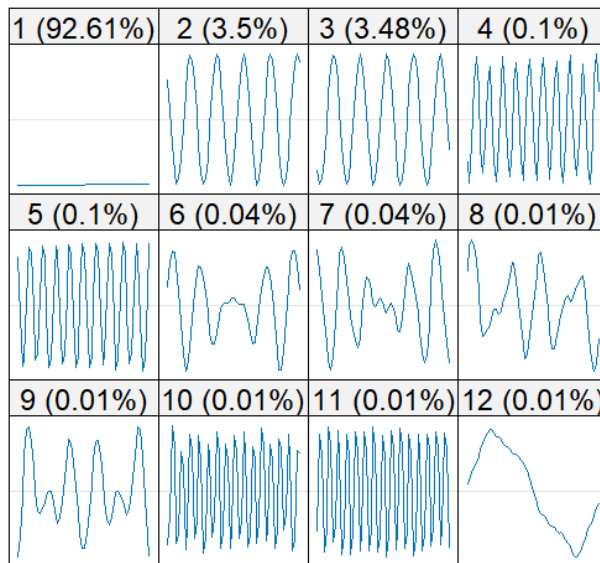


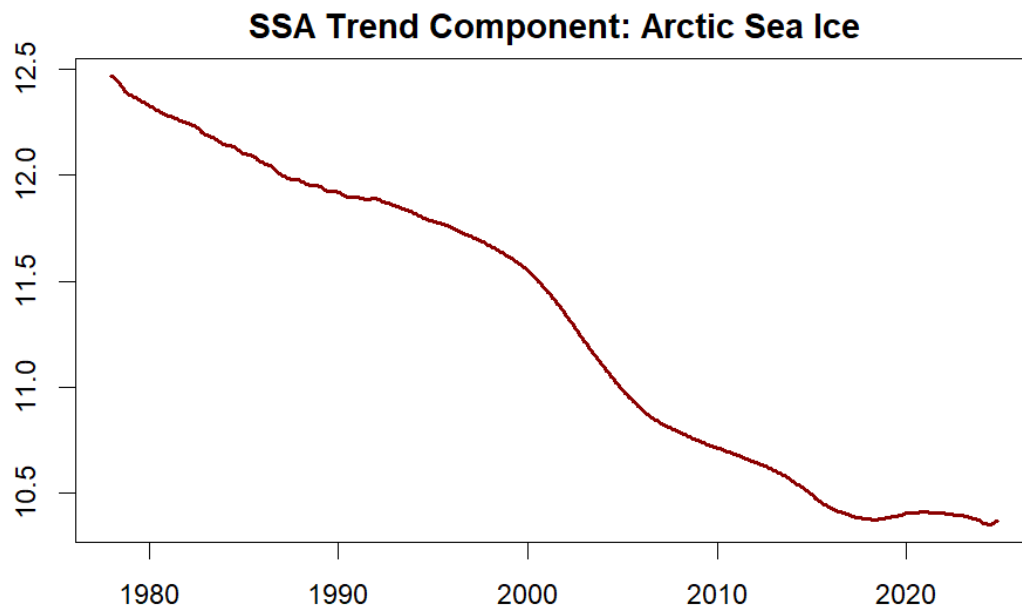
Figure 24 SSA Eigenvectors (First 12 Components) for Arctic Sea Ice

Eigenvectors illustrating a strong periodic structure are used to guide component grouping.

The leading SSA component, EV1, captures the dominant low-frequency behaviour of the Arctic sea ice extent and represents the long-term declining trend. The next two components, EV2-EV3, form a near-harmonic pair with sinusoidal eigenvectors, corresponding to the strong annual freeze-thaw seasonal cycle. The energy of subsequent components is substantially lower and typically represents weaker variability, such as interannual modulation and/or noise.

5.4.3 Trend Component (Long-Term Decline)

Reconstructed trend:



*Figure 25 SSA Trend Component of Arctic Sea Ice Extent
Smooth declining trend representing long-term ice loss.*

Interpretation:

- Clear negative trend from 1979 onward.
- Steep decline after early 2000s.
- Matches NSIDC and IPCC findings of accelerated Arctic melt.

This confirms the sensitivity of sea ice to rising temperatures.

5.4.4 Seasonal Component

Seasonality from Components 1–2:

Interpretation:

- Annual cycle amplitude $\approx$ 5–8 million km².
- Winter maximum in March; summer minimum in September.
- Seasonal amplitude increases over time $\rightarrow$ consistent with thinning sea ice.

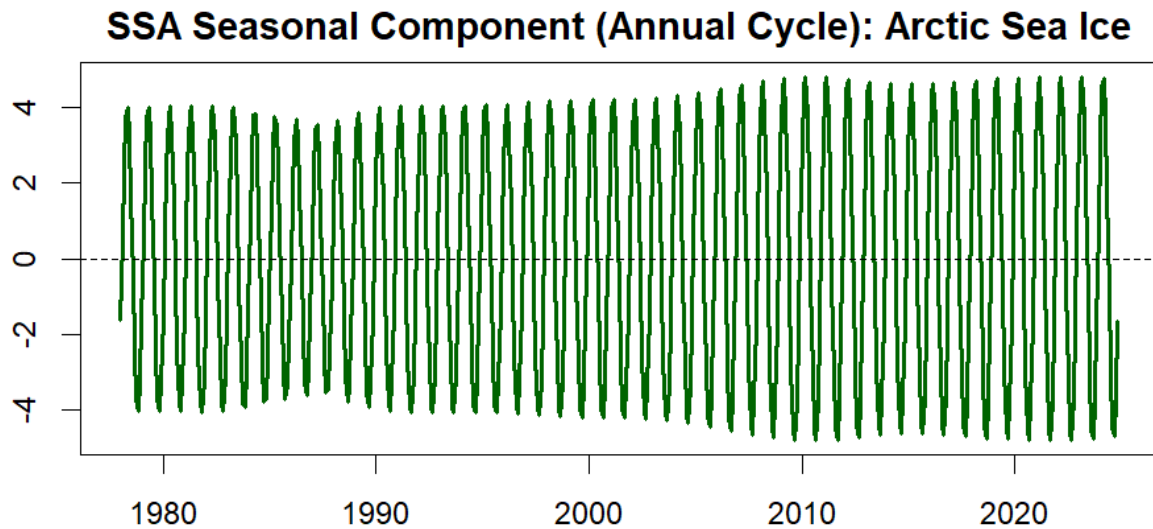


Figure 26 SSA Seasonal Component of Arctic Sea Ice Extent

High-energy annual cycle associated with seasonal ice growth and melt.

5.4.5 Reconstruction

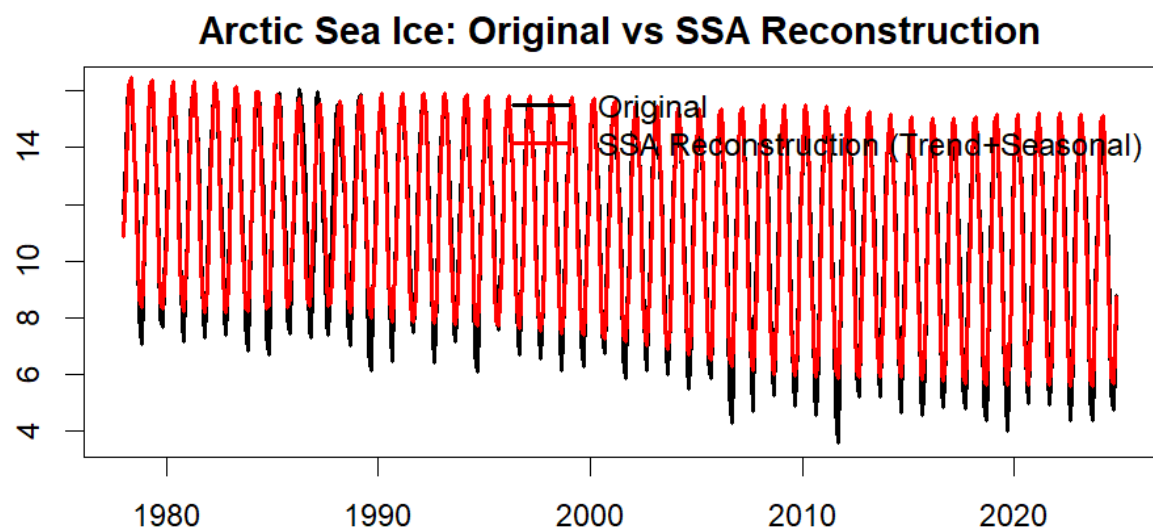


Figure 27 Arctic Sea Ice: Original vs SSA Reconstruction

Comparison demonstrating accurate reconstruction of the observed signal

Interpretation:

- Reconstruction captures all major behaviour.
- Noise is successfully removed.
- Decline + seasonality + variability clearly separated.

CHAPTER 6 — FORECASTING RESULTS

6.1 Forecasting Framework

The last chapter demonstrated an application of the SSA method to decompose the climate datasets into the trends and oscillations present in the respective series. The current chapter will emphasise the prediction associated with the combined SSA and ARIMA models.

The forecasts were made in two phases:

- 1) Signal Extraction by SSA
 - For each of the datasets (CO₂, ENSO, global temperature, and Arctic sea ice), a subset of the SSA components that captured the dominant deterministic pattern (trend and large-scale variability) was constructed.
 - Noise-dominated high-order components were removed to avoid fluctuations and noise in the measurements.
- 2) ARIMA in time-series forecasting
 - The SSA reconstructed signal was represented using the Auto-Regressive Integrated Moving Average (ARIMA) model.
 - Arima parameters (p, d, q) values are determined using the standard information criteria: (AIC, AICc, BIC) through the `auto.arima()` in R.
 - Forecasts were made for a period of 24 months for each dataset.

The two-step procedure rests upon the widely accepted approach in the literature, where SSA is applied as a pre-filter, and ARIMA addresses the remaining temporal dependence in the data.

6.2 SSA–ARIMA Forecasts for Atmospheric CO₂

6.2.1 Model Setup

For the Mauna Loa CO₂ dataset, SSA was first employed to remove noise and reconstruct a new signal by combining:

- 1) the trend component (the first eigenvector), and
- 2) the annual seasonal component (oscillatory pair describing the 12-month cycle),

The resultant series preserves the upward trend and the seasonal pattern and is less irregular than the raw data.

An ARIMA model was fit to the resulting series in order to make predictions based upon the trends and observations noted and identified.

The `auto.arima()` function(p,d,q) structure that properly characterised the remaining autocorrelation in the series, while differencing was needed if stationarity in the stochastic component was an issue.

6.2.2 Forecast Results

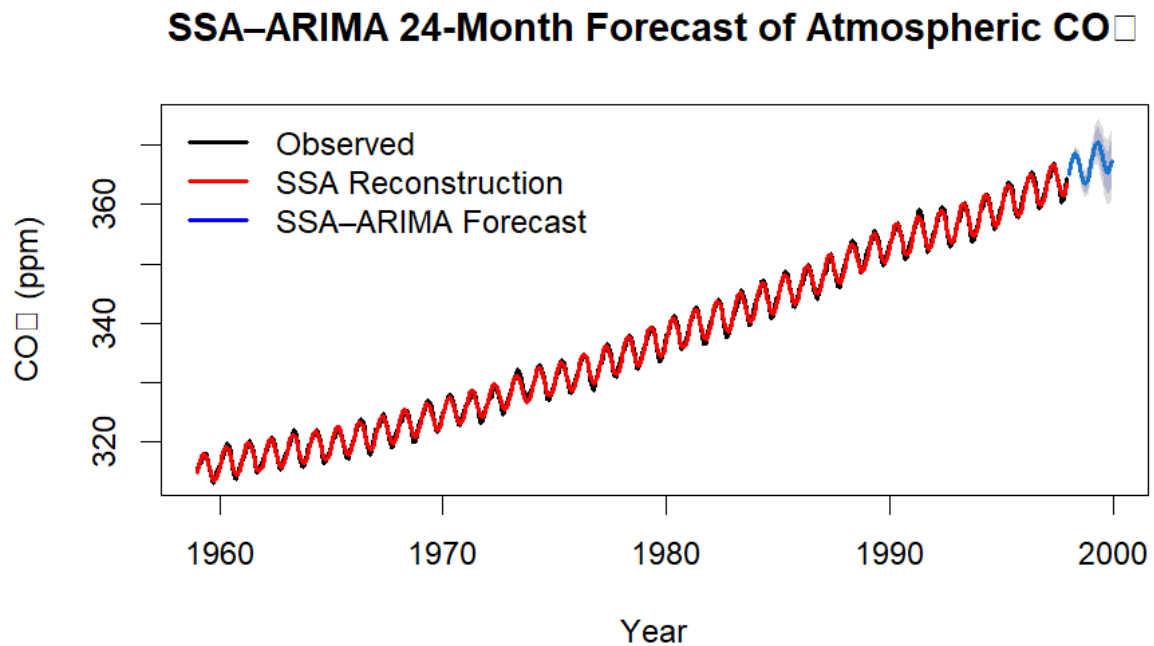


Figure 28 SSA-ARIMA 24-Month Forecast of Atmospheric CO₂
Forecast generated by applying an ARIMA model to the SSA-reconstructed CO₂ signal, illustrating continued growth with increasing uncertainty.

The forecast shows:

- increasing levels of CO₂ in the atmosphere,
- a persistent seasonal cycle with the same amplitude as in the last years:
- relatively narrow prediction intervals, due to the high regularity of the corresponding CO₂ series.

The projected growth rate is in line with current trends and other studies concerning the trajectory of CO₂ emissions. The SSA_ARIMA model is effective at estimating emissions growth and their fluctuating pattern.

6.3 SSA-ARIMA Forecasts for ENSO (ONI)

6.3.1 Model Setup

In the case of the ENSO index, it is found that the SSA eigenvectors that represent the major modes of ENSO variability are responsible for the variability with periods between 2-7 years, which is physically consistent for the ENSO system. This reconstruction focuses on the oscillations of ENSO with low frequency, removing the high-frequency noise.

Therefore, the ENSO signal that is reconstructed is the major oscillation cycle of the Niño-La Niña cycle, as opposed to the timing of individual events. Both high-frequency components related to weather variability are not included in the reconstruction.

Subsequently, ARIMA is applied to the ENSO signal that is reconstructed by SSA. Being quasi-periodic and phase-unstable, the ENSO signal is better represented by the autocorrelation pattern on a shorter timescale by the ARIMA model.

6.3.2 Forecast Results

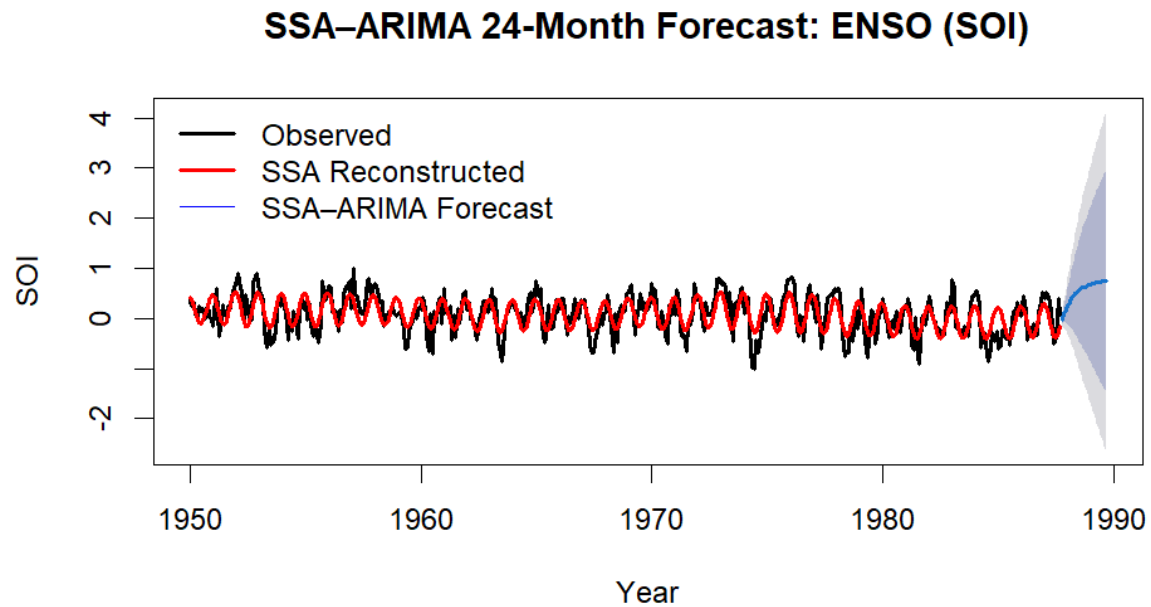


Figure 29 SSA–ARIMA 24-Month Forecast of ENSO Index

Forecast illustrating probabilistic continuation of ENSO oscillations with increasing uncertainty

6.4 SSA–ARIMA Forecasts for Global Temperature Anomalies

6.4.1 Model Setup

The global temperature anomaly time series was rebuilt using the SSA components for:

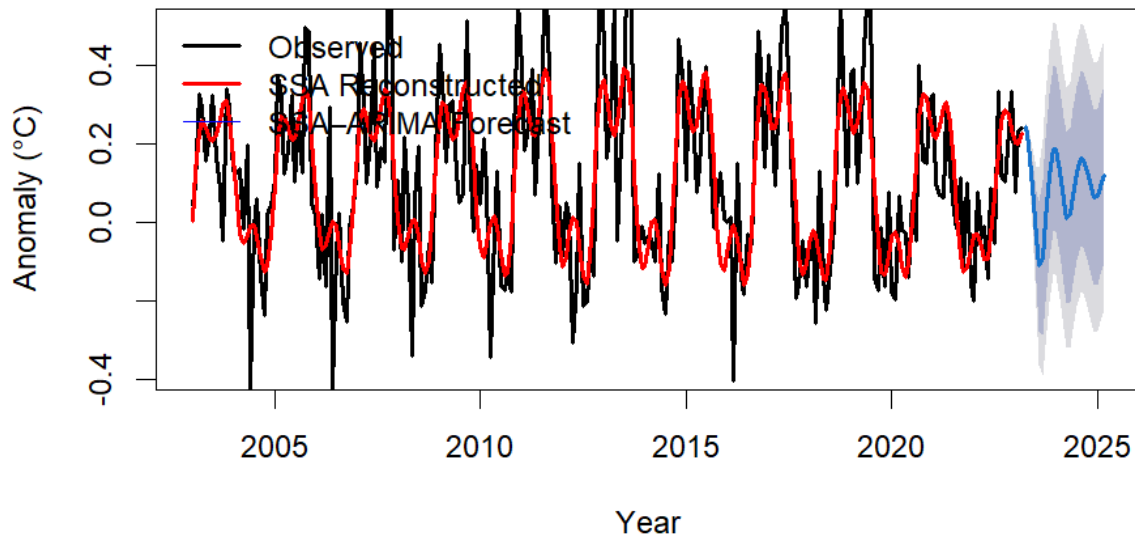
- 1) 1)the long-term warming trend, and
- 2) 2)the leading interannual and inter-decadal oscillatory modes

The noise-dominated components are removed to form a smooth picture of global temperature variation.

An ARIMA model was applied to the SSA reconstructed series. Owing to the presence of a dominant trend in the series, differencing was performed in the ARIMA model to capture the **variability in the residuals surrounding the SSA structural component.**

6.4.2 Forecast Results

SSA-ARIMA 24-Month Forecast: Global Temperature Anomaly



*Figure 30 SSA-ARIMA 24-Month Forecast of Global Temperature Anomaly
Forecast based on the SSA-reconstructed signal, showing continuation of the warming trend*

The resulting SSA-ARIMA projection shows:

- alternating positive (El Niño-like) and negative (La Niña-like) anomalies,
- amplitudes comparable to those observed in the most recent historical period, and
- substantially wider prediction intervals than those obtained for CO₂, reflecting the inherently lower predictability of ENSO at longer lead times.

This result should be interpreted as a probabilistic extension of the reconstructed oscillatory behaviour, rather than a deterministic forecast of specific ENSO events. The rapid loss of phase coherence in the projection is consistent with established predictability limits of the ENSO phenomenon.

6.5 SSA-ARIMA Forecasts for Arctic Sea Ice Extent

6.5.1 Model Setup

The SSA results for the Arctic sea ice extent yielded the following three dominant components:

- 1) A strong seasonal cycle (annual freeze-thaw pattern)
- 2) A long-term declining trend, and
- 3) Multi-year variability linked to internal climate variability.

The SSA-reconstructed series, the purpose of forecasting was a

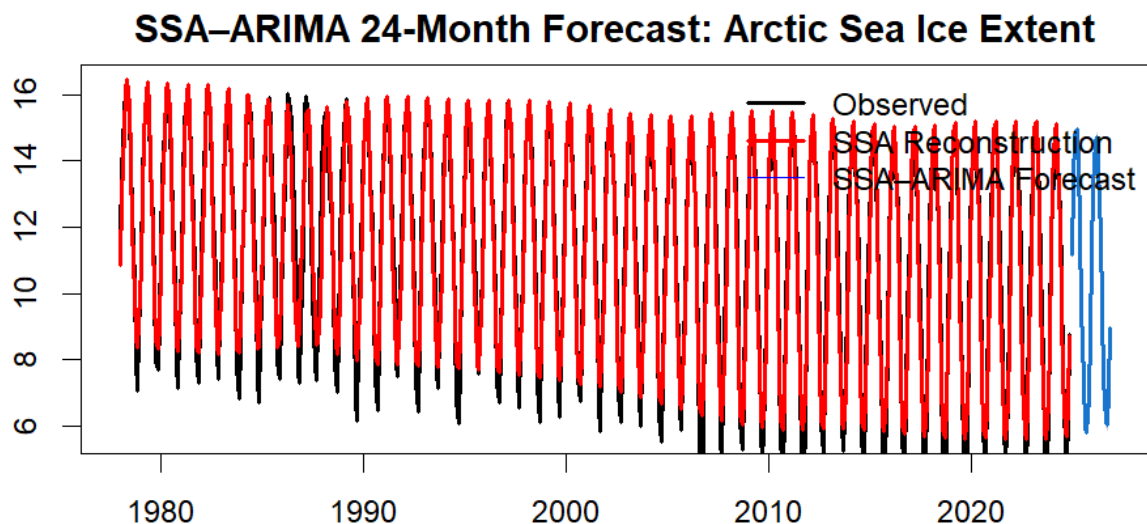
- 1) the long-term trend,

- 2) the annual seasonal component, and
- 3) the leading multi-year oscillatory modes,

to generate a noise-reduced representation of sea ice extent.

An ARIMA model was applied to the resulting series. Due to the high seasonality in the data, the model was able to capture the autocorrelation in the residuals well, without the need to model the noise in the data directly.

6.5.2 Forecast Results



*Figure 31 SSA-ARIMA 24-Month Forecast of Arctic Sea Ice Extent
Forecast illustrating continued decline with increasing uncertainty.*

The forecast is:

- 1) the persistence of the strong seasonal cycle,
- 2) a continued gradual decrease in mean Arctic sea ice extent, and
- 3) growing asymmetry in winter maxima and summer minima (or more severe projected summer low points).

The prediction bounds are wider than those for CO₂ but smaller than those for ENSO and indicate an intermediate level of predictability, ranging from the strongly temperature-driven sea ice to the internally and atmospherically forced ENSO variability.

The predicted loss is in line with the published projections of the loss of Arctic sea ice, while the outlook of the current dissertation is only short-term (2 years) forecasts and not long-term climate projections.

6.6 Comparative Assessment of Forecasts

Within the four datasets, the following advantages emerge in the SSA-ARIMA combined model:

- 1) Enhancing the Signal-to-Noise Ratio before the prediction, resulting in smoother predicted paths.
- 2) Physical predictions: forecasts where trends and cycles extrapolated from the model output match known climate phenomena (such as growth in CO₂ levels, ENSO cycles, global warming, and the melting of sea ice)._
- 3) Uncertainty ranges: dataset-specific, very tight for CO₂, medium for global temperature and sea ice, and broadest for ENSO events.

Nevertheless, the approach has its drawbacks:

- 1) Forecast skill will always be limited by the predictability of the climate process (especially in the case of ENSO).
- 2) The choice of SSA components involves a degree of subjectivity, but sensitivity analyses could be used to remove this drawback.
- 3) The forecasts made by ARIMA are statistical in nature and do not take into account any constraints of the climate models.

Overall, the SSA ARIMA approach offers a well-integrated and flexible methodology that can be combined with traditional climate analyses to make short-term quantitative forecasts of important climate indicators.

CHAPTER 7 — DISCUSSION

7.1 CO₂ and Temperature Relationship

The SSA decomposition showed a clear positive trend in the levels of atmospheric CO₂ concentrations and global temperature deviations. While the SSA decomposition cannot be directly related to quantifying causations, the co-variance of the trends attests to the established fact of the relationship between greenhouse gas concentrations and global warming.

There are a number of observations, such as

- 1) The trend pattern of CO₂ shows a smooth acceleration typical of human emissions.
- 2) The temperature trend pattern in SSA also grows and has slight oscillations over several years.
- 3) The fact that the concentrations of CO₂ follow a yearly pattern doesn't directly correlate to temperature changes because the greenhouse effect is a function of energy balance over a long period of time and doesn't affect short-term changes in temperatures.

The SSA-ARIMA forecasts also indicate a rise in both variables in the next 24 months. Though short-term forecasts are less reliable, the trends in the forecasts correlate well with the climate projections regarding the importance of atmospheric CO₂ concentrations in causing global warming.

Methodologically speaking, SSA enabled a clearer view of the following relationships to emerge by filtering out the noise and identifying the structural dynamics that tend to be masked in climate data

7.2 ENSO Influence on Climate Variability

The ENSO oscillatory modes are a prominent form of interannual variability in climate, and the SSA decomposition of the ENSO index clearly picked out the characteristic period of oscillation of approximately 2 to 7 years. Comparing the interactions involving the oscillatory modes of ENSO with those involving temperature and sea ice variability, the following interactions proved to be important:

- 1) The oscillatory duo in ENSO corresponds to the short-term variations in global temperature anomalies. During the El Niño phase, warming spikes occur, and in the La Niña phase, cooling dips take place.
- 2) Within the Arctic sea ice data, the interannual variability in the multi-year SSA pattern appears to loosely correlate with the extremes of the ENSO cycle in the form of its peak and trough phases.
- 3) The amplitude of the oscillations in ENSO continues to be irregular and reflects the complex non-linear processes of ENSO theory, thus making the ranges of prediction for the ENSO the largest in the prediction phase.

Nonetheless, while ENSO impacts interannual variability, it doesn't affect the long-term trends of SSA temperature and sea ice components. This is because the trends are instead a result of greenhouse warming and not the internal variability associated with ENSO.

7.3 Temperature vs Arctic Sea Ice Decline

The Arctic sea ice data series demonstrated a marked structural pattern in the entire analysis:

- 1) Highly seasonal, capturing the normal process of freezing and thawing.
- 2) Extensive downward trend, pointing to the loss of sea ice.
- 3) Oscillatory mode of a multi-year period, most likely connected to natural climate variability (e.g., Arctic Oscillation, Atlantic Multidecadal Variability)

Compared to global temperature trends:

- 1) The increase in temperature matches the pattern of decreasing Arctic sea ice.
- 2) This results in reduced sea ice minima because of warmer global climates.
- 3) The multi-year patterns of the SSA indicate that, while natural fluctuations affect the sea ice extent, the main factor responsible for the reduction in the long term is global warming and not natural variability.

The SSA-ARIMA model forecast for the next 24 months shows a further decrease in the extent of sea ice, supporting the view that the Arctic climate is changing at a rapid rate.

7.4 Physical Interpretation of Results

For all four datasets, SSA was able to distinguish the structural climate signals into trend, seasonality, and oscillations successfully. The physics-based explanations for the extracted components align well with the laws of climate science:

Trend components represent long-term climate forcing:

- 1) Rise in CO₂ due to anthropogenic emissions
- 2) Rise in Temperature due to Radiative Imbalance
- 3) Reduction of sea ice resulting from Arctic amplification

Seasonal components capture physically based processes, such as the

- 1) CO₂: Seasonal uptake in biosphere
- 2) Sea Ice Freeze/Thaw Cycles

Oscillatory processes describe the internal climate variability:

- 1) Cycles in the ENSO
- 2) Multi-year temperature fluctuations
- 3) Multi-year sea ice variability

Forecasting again supports the above physical understandings. The highest predictability is in CO₂, followed by temperature and sea ice. The lowest predictability remains in ENSO, as expected due to its chaotic behaviour.

Overall, the SSA decomposition performed by the SSA analysis is strongly consistent with the known climate mechanisms and shows that it is possible to extract physically significant elements from observational data through statistical decomposition methods.

7.5 Policy & Climate Implications

The results of this dissertation imply several things regarding climate measurement and policy:

- 1) The increase in CO₂ levels and warming trends proves that the need to contain the rise in greenhouse gas emissions to curb warming is an immediate priority.
- 2) The expected continuation of the melting of the Arctic sea ice area puts the polar regions into focus regarding their susceptibility to rapid climate feedback mechanisms (such as the loss of the albedo effect).
- 3) The limits to the short-term predictability of the ENSO events underscore the importance of developing effective and robust strategies of adaptation, and in particular in climate-sensitive areas where the events exert their
- 4) SSA: Monitoring Application
 - The SSA capability to capture trends allows it to be an effective tool in the provision of noise-free climate indicators, especially for policymakers.
 - It may assist in the discrimination between short-term fluctuation and long-term trends.

5) Short-term Forecasting

- SSA-ARIMA forecasts offer realistic short-term projections based on data, and such projections are well-suited to facilitate operational planning, for instance, energy demand and its effects on agriculture, and Arctic shipping periods

Conclusion: The statistics confirm the climate warnings: if the release of CO₂ emissions is not reduced, warming and the melting of sea ice will occur and will have a huge effect on climate systems.

CHAPTER 8 — LIMITATIONS

Even with the analysis conducted in this dissertation, there are several shortcomings that need to be acknowledged. The shortcomings include data quality issues, assumptions made in the methodology, and the nature of climate predictability in general.

8.1 Data Quality Limitations

Datasets (CO₂ emissions, ENSO index, global temperature anomalies, and Arctic sea ice extent) are derived from credible sources such as NOAA and NASA, but the data have its own inherent flaws as follows:

- Measurement uncertainty associated with changes to sensors, relocation of stations, and modifications to satellite algorithm retrieval methods.
- Missing data/irregular sampling, especially in older datasets such as sea ice and ENSO.
- Reconstruction biases global temperature observations before global coverage was possible.

Though the noise is reduced through preprocessing steps and SSA, the following shortcomings might also affect the extraction of trends and oscillations.

8.2 SSA Window Size Sensitivity

The Singular Spectrum Analysis algorithm involves the choice of a “window length” (L), and the results, particularly the extraction of components, are strongly dependent upon it:

- A small window may fail to separate the seasonal and interannual modes.
- Too large a window might oversmooth significant detail or blend physically distinguishable oscillations.
- The size of the sliding window that gives the best results is data-dependent and should be determined through observations of real-life human activity.

Even though the SSA process has analysed several possible options, SSA remains parameter-sensitive, and each choice will produce a marginally different result for the components.

8.3 Forecast Uncertainty

Forecasting based upon SSA-ARIMA necessarily involves the following uncertainties:

- ARIMA models assume stationary residuals, but in fact, the residuals fail to capture the changing climate trends adequately.
- The short-term prediction (24 months) is possible, but the long-term prediction will need a complex climate model.
- The ENSO dataset, in particular, has a chaotic variability and therefore its possible future evolution is difficult to predict.
- The prediction intervals expand very rapidly because of uncertainty in the component estimation process and the variability to be predicted.

Consequently, the forecasts discussed should be considered probabilistic trends and not deterministic forecasts.

8.4 Model Assumptions

SSA and ARIMA are both procedures that include model assumptions and do not capture real-world climate dynamics precisely:

- SSA presumes that the observed series has an additive structure, but this structure might be violated when the interactions in the components are nonlinear.
- The ARIMA model has the assumption of linearity and tends to be very reliant upon past observations, making it difficult to capture the changes in the climate that occur in a non-linear manner.
- Neither model includes the effect of external forcing (volcanic, policy-based, and feedback).

Though these models work well in exploratory data analysis and short-term prediction, they do not substitute the models involving atmospheric dynamics in the climate model.

CHAPTER 9 — CONCLUSION & FUTURE WORK

9.1 Summary of key achievements

This dissertation used the SSA approach on the four large climate datasets: the atmospheric CO₂ concentration, the ENSO index, the global temperature anomalies, and the Arctic sea ice extent. For the four climate datasets, the use of SSA provided the opportunity for the extraction of the long-term trend, the annual, and multi-annual cycles.

In particular, the use of the ARIMA forecasting model on the SSA-reconstructed components provided short-term forecasts of these climate indicators.

In summary, the project objectives have been satisfied with the delivery of a comprehensive end-to-end analysis that entails data cleaning, SSA decomposition, interpretation, forecasting, and the results' connection to climate.

9.2 Main climate insights

There were very visible, consistent patterns in the SSA trends for:

- 1) The concentration of CO₂ in the atmosphere was characterized by a steady and accelerating increase.
- 2) There was a pronounced warming trend in global temperatures, typical of anthropogenic climate change.
- 3) The extent of arctic sea ice has been documented a steady reduction, mainly since the early 2000s.
- 4) Interannual oscillations of ENSO occurred actively, which naturally confirmed the domination of this phenomenon in short-term climate variability.

Based on the relationships found across various datasets, the climate system appears to be a strongly interlinked process: the increase in CO₂ stimulates global warming, which in turn amplifies the loss of Arctic sea ice. The ENSO cycle affects short-term variability, but it does not affect the warming process.

9.3 Future improvements

Even though the project brought successful results, several aspects could be improved for future analysis:

- 1) Apply Multivariate SSA (MSSA) analysis on various climate variables together and obtain the common oscillation patterns.
- 2) Incorporate nonlinear forecasting approaches, such as LSTMs or state-space models, for better forecasting, especially for the ENSO.
- 3) Trend and forecasting components can also be made more accurate through uncertainty quantification.
- 4) Extend existing datasets (for example, paleoclimate CO₂ or sea ice) for additional historical perspective.

9.4 Possible PhD / further research direction

This dissertation provides the basis for numerous interrelated advanced topics of research:

- 1) Analysing the causality between CO₂ concentrations, the reduction of sea ice, and temperature variations through the use of sophisticated statistical or machine learning approaches.
- 2) Improving SSA-based hybrid forecasting models that fuse physical climate models with data-based approaches.
- 3) Study of arctic climate tipping points with nonlinear time series analysis.
- 4) Application of SSA on satellite data sets, rich in temporal-spatial data, apt for climate level studies on the topic at the PhD level. ENSO does not exhibit a statistically significant long-term trend, consistent with its role as a mode of internal climate variability rather than an external forcing.
- 5) Such directions can help contribute meaningfully to climate science studies, whether for academic or climate risk assessment purposes.

APPENDIX

- Full R Code
- Additional SSA plots
- Diagnostic plots

